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(71) Applicant: **Edwards Lifesciences Innovation (Israel) Ltd.**, Caesarea (IL)

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(72) Inventors: **Yuval Zipory**, Modi'in (IL); **Alexei Koifman**, Melbourne (AU); **Ehud Aviv**, Costa Mesa, CA (US); **Tal Reich**, Moledet (IL)

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(73) Assignee: **Edwards Lifesciences Innovation (Israel) Ltd.**, Caesarea (IL)

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(74) *Attorney, Agent, or Firm* — Anya Adams

Related U.S. Application Data

(63) Continuation of application No. 17/378,559, filed on Jul. 16, 2021, now Pat. No. 11,766,263, which is a (Continued)

(57) **ABSTRACT**

(51) **Int. Cl.**

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An anchor-handling device comprises a housing shaped to define a channel having an anchor-storage zone and a proximal opening. A tissue anchor is stored in the anchor-storage zone. The anchor-handling device is configured such that, while the tissue anchor is stored in the anchor-storage zone, the tissue anchor is movable out of the anchor-storage zone toward the proximal opening in response to a proximally-directed force being applied to the tissue anchor. The anchor-handling device comprises an element that serves as an indicator of movement of the tissue anchor out of the anchor-storage zone toward the proximal opening by the element moving in response to the proximally-directed force. Other embodiments are also described.

(52) **U.S. Cl.**

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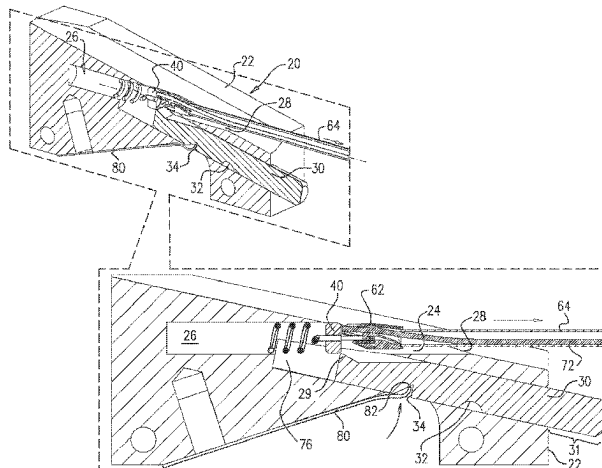
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(58) **Field of Classification Search**

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23 Claims, 18 Drawing Sheets



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(58) **Field of Classification Search**

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See application file for complete search history.

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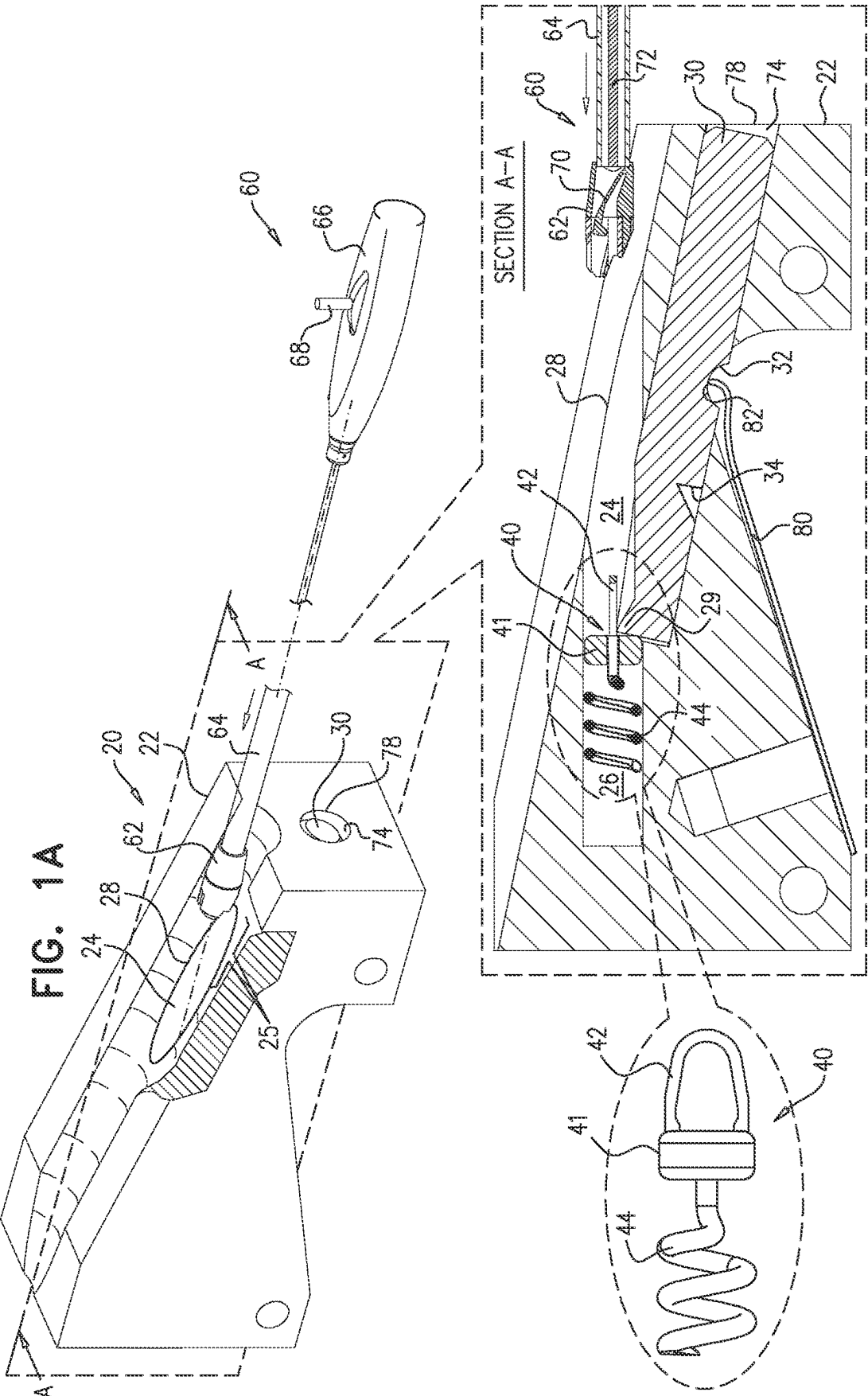
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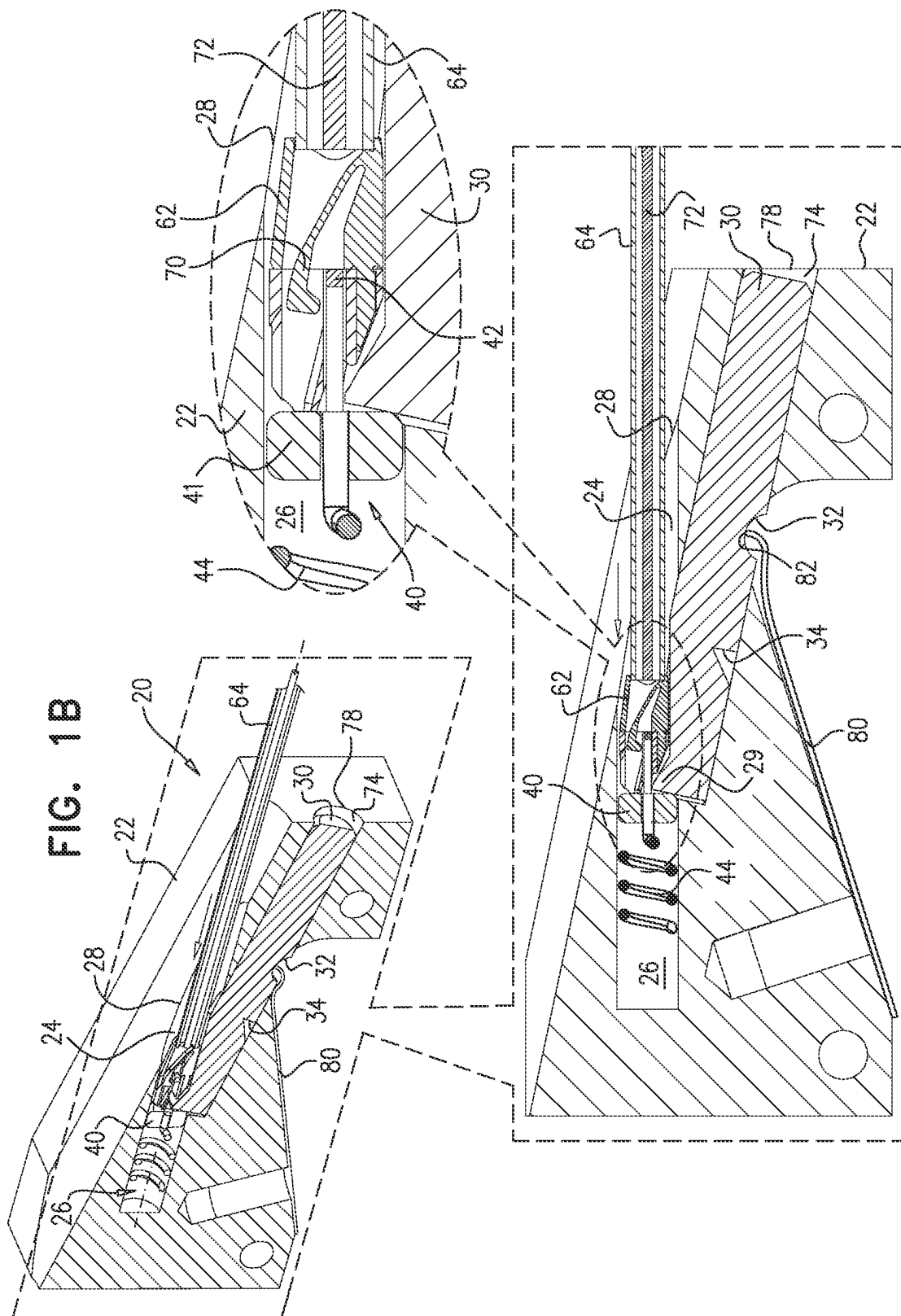
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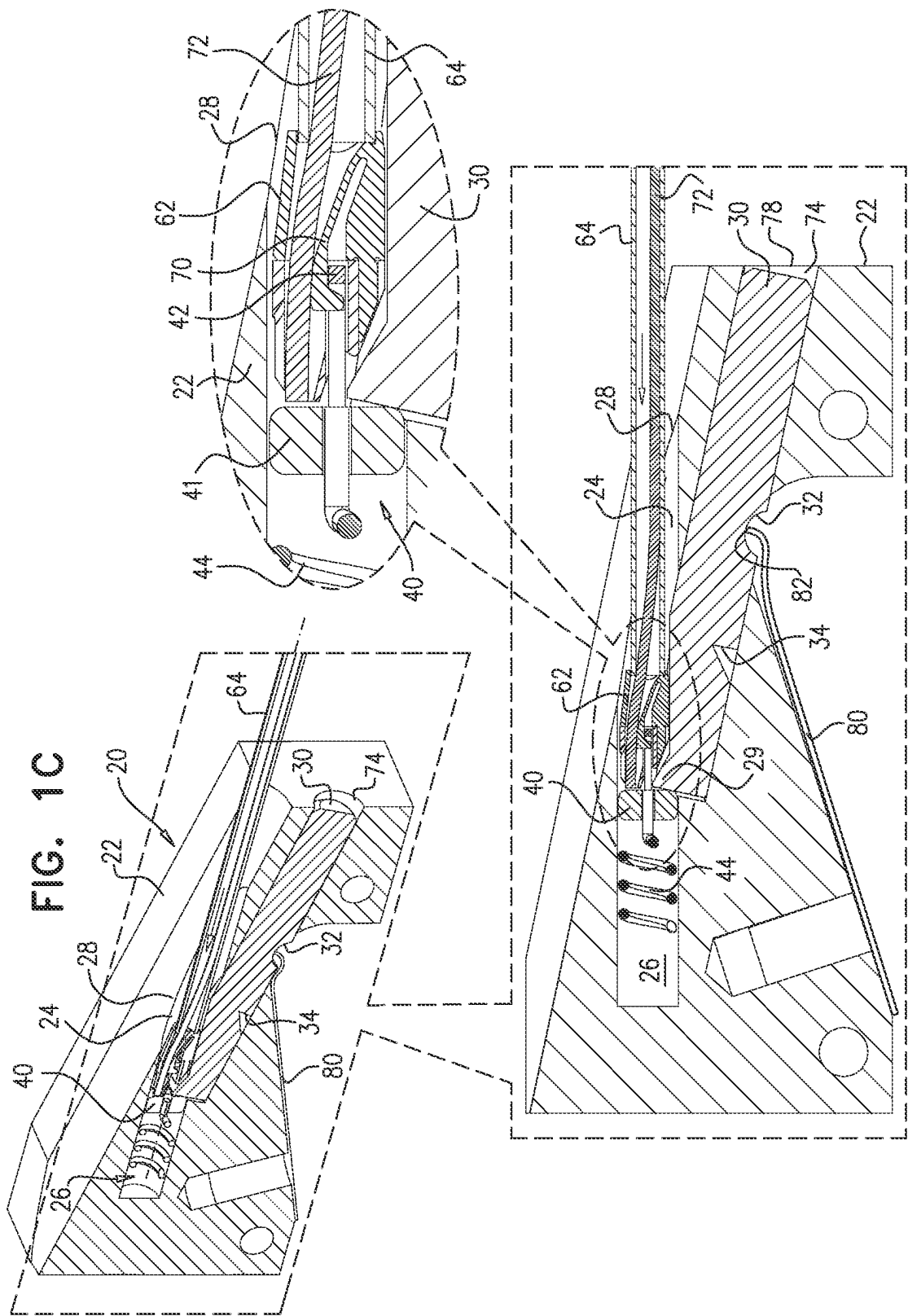
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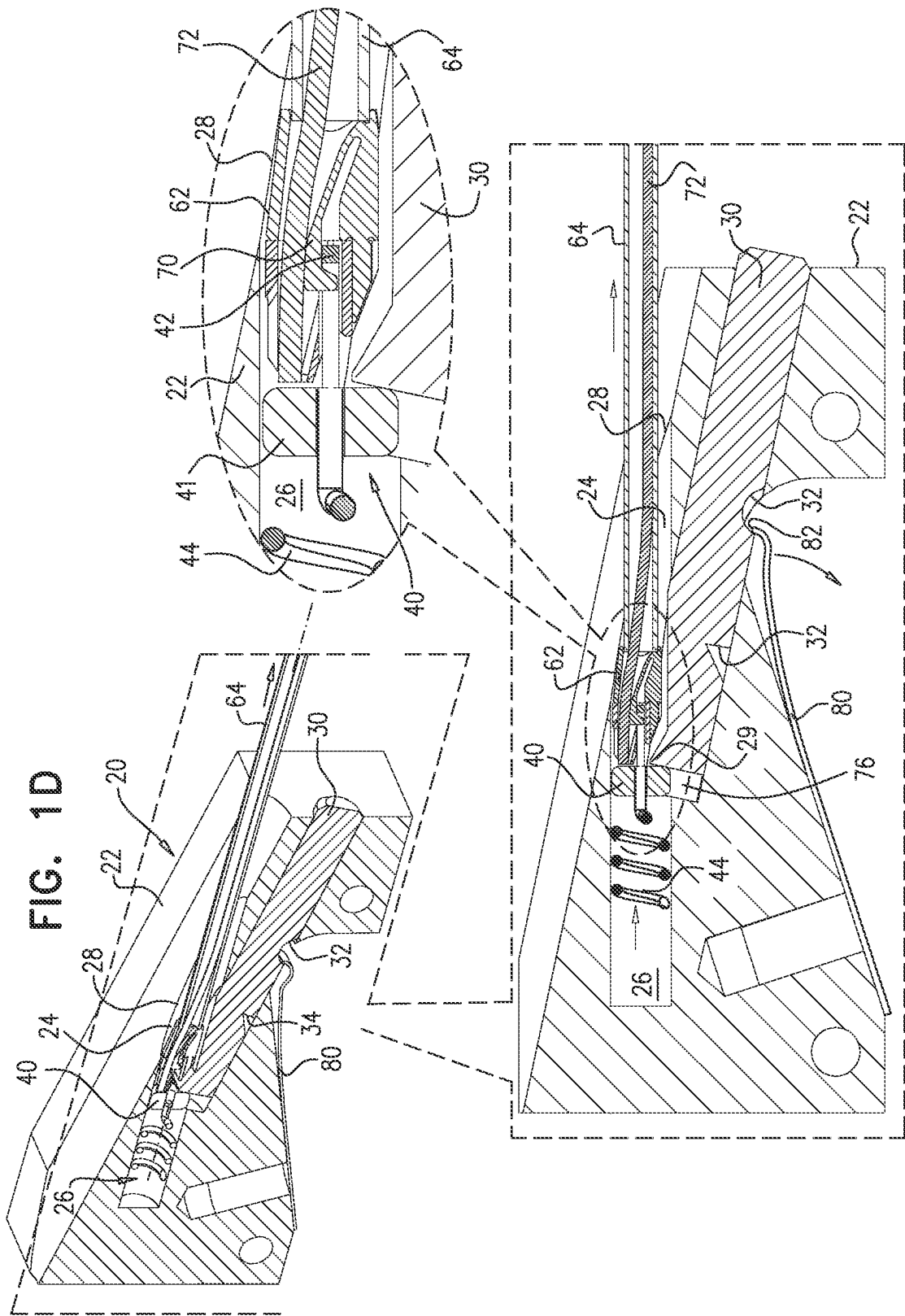
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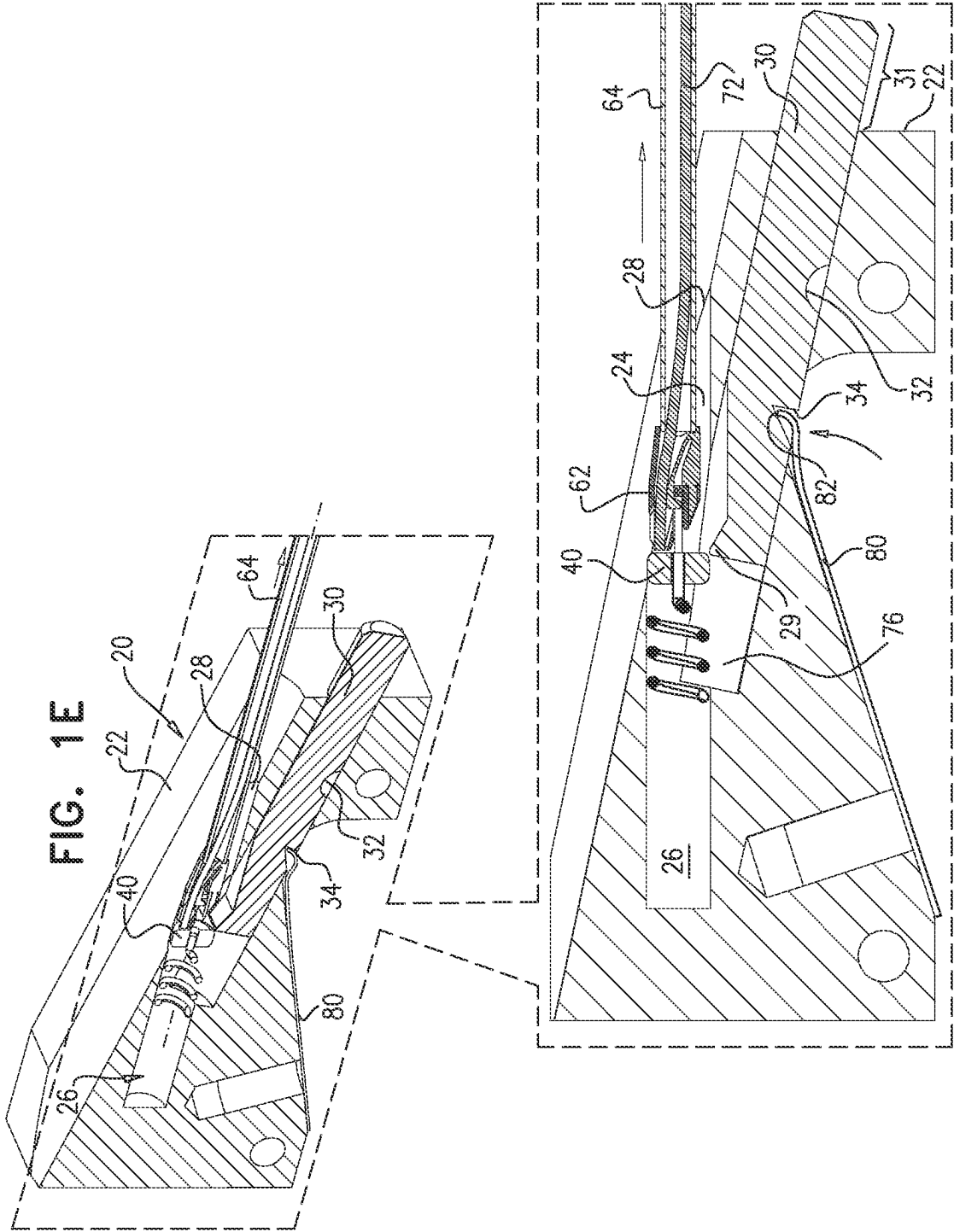
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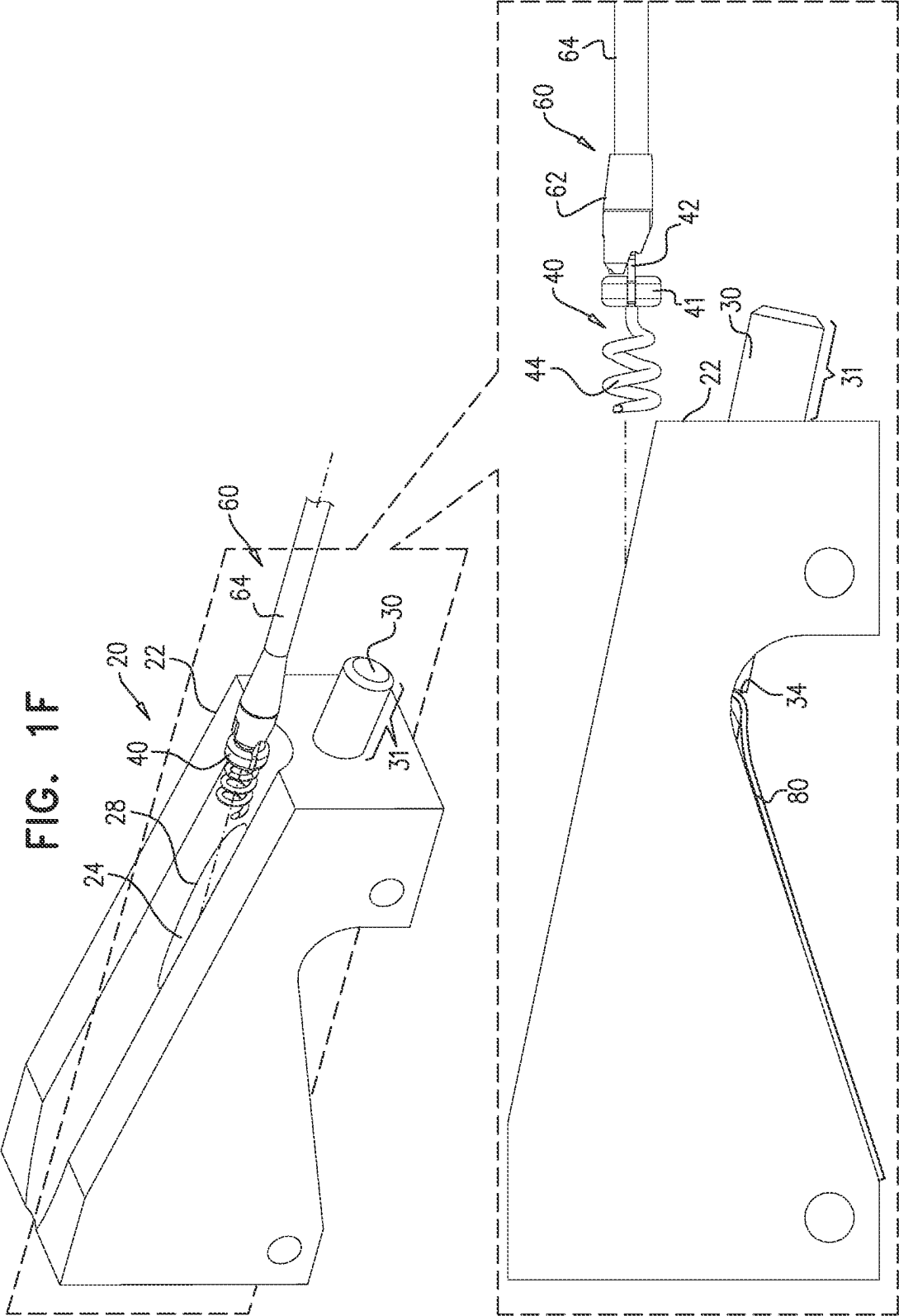












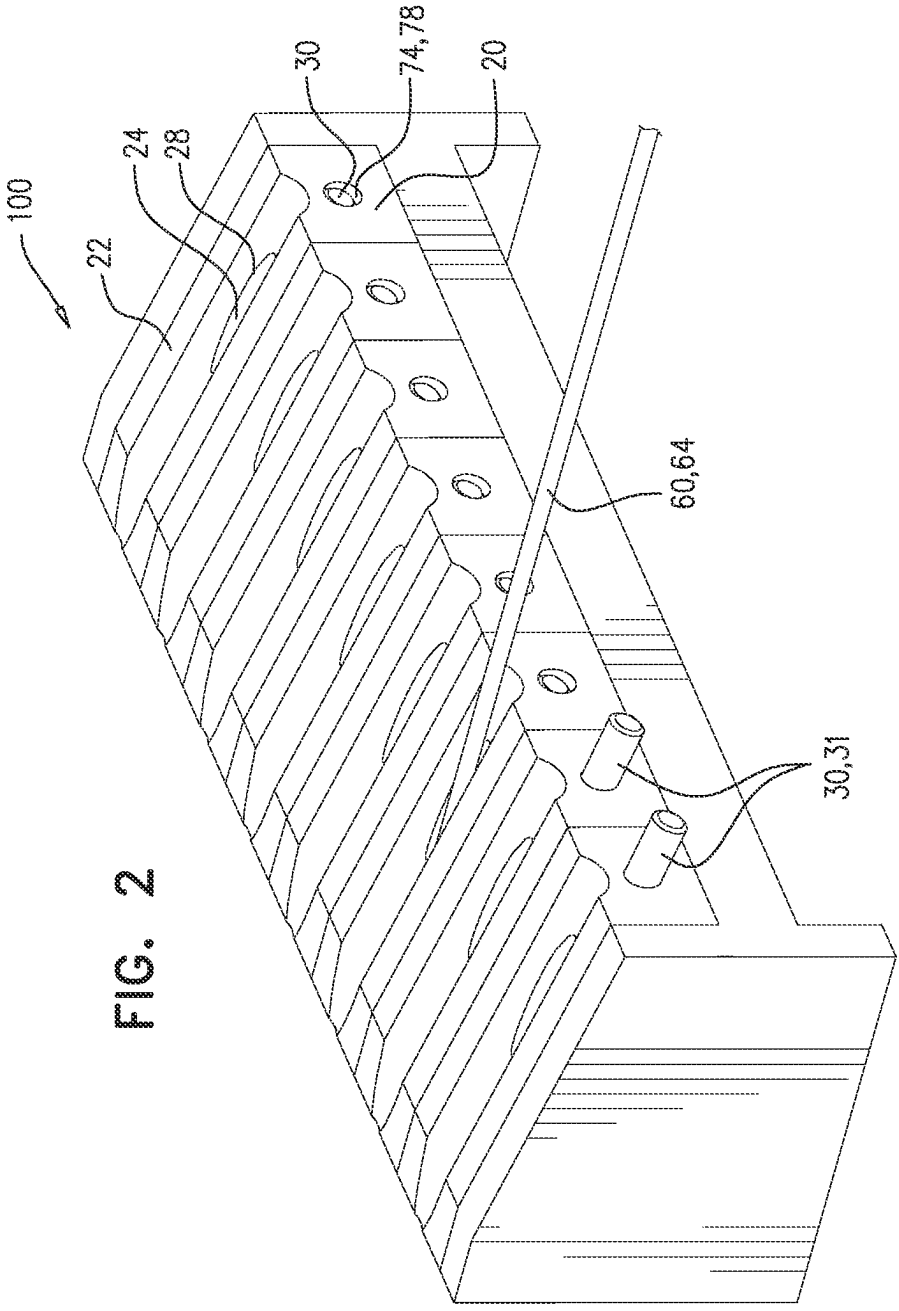


FIG. 3A

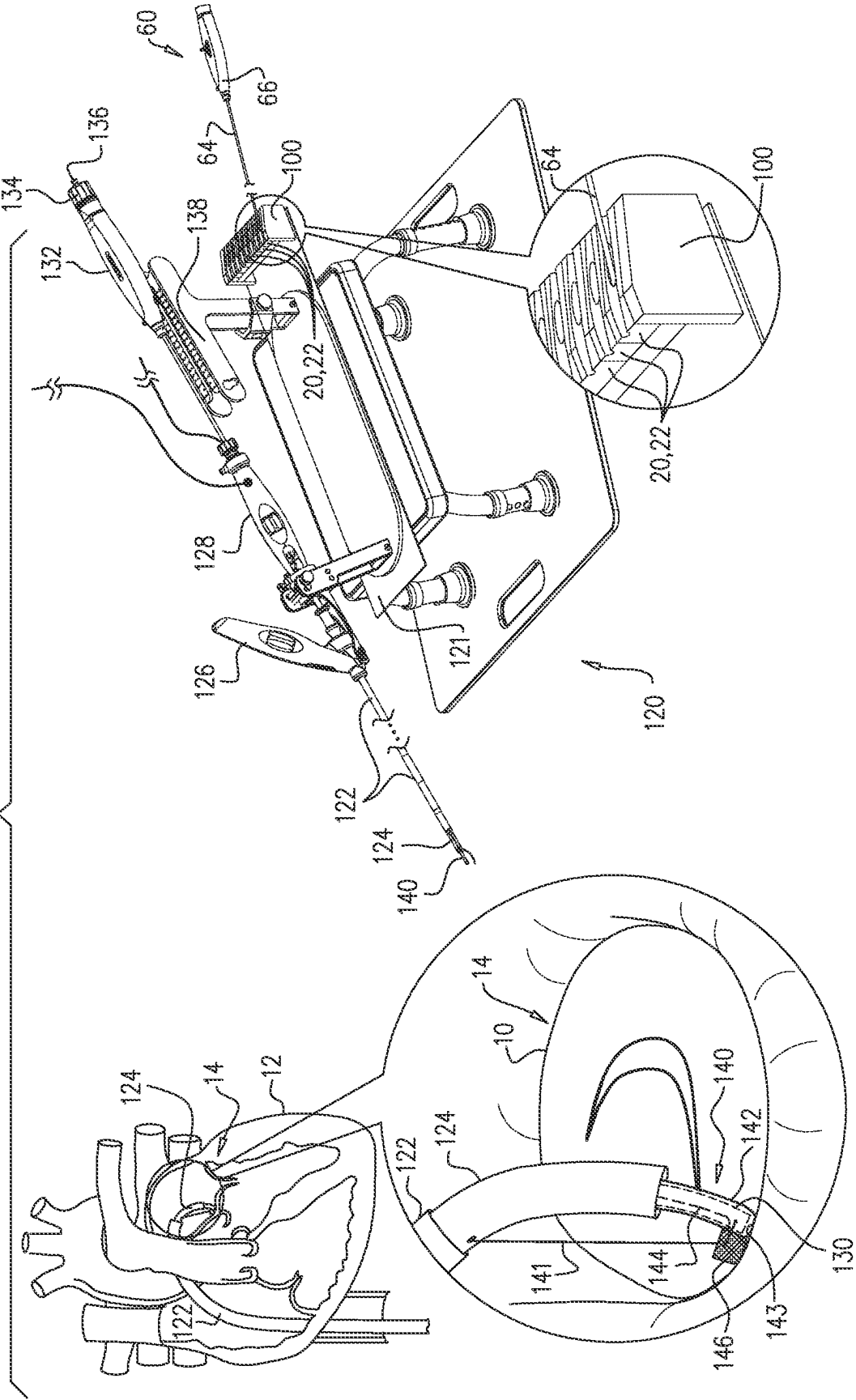
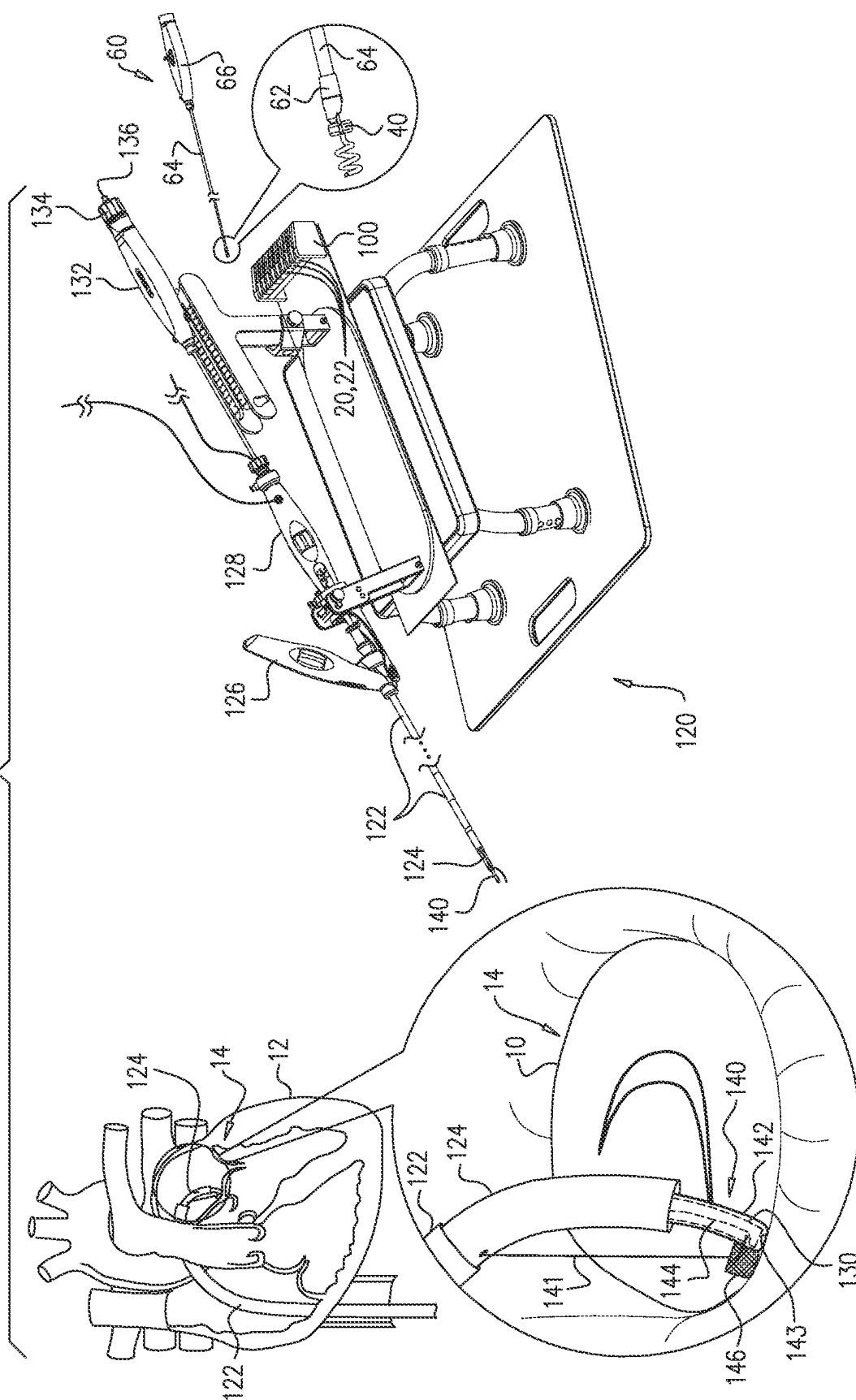


FIG. 3B



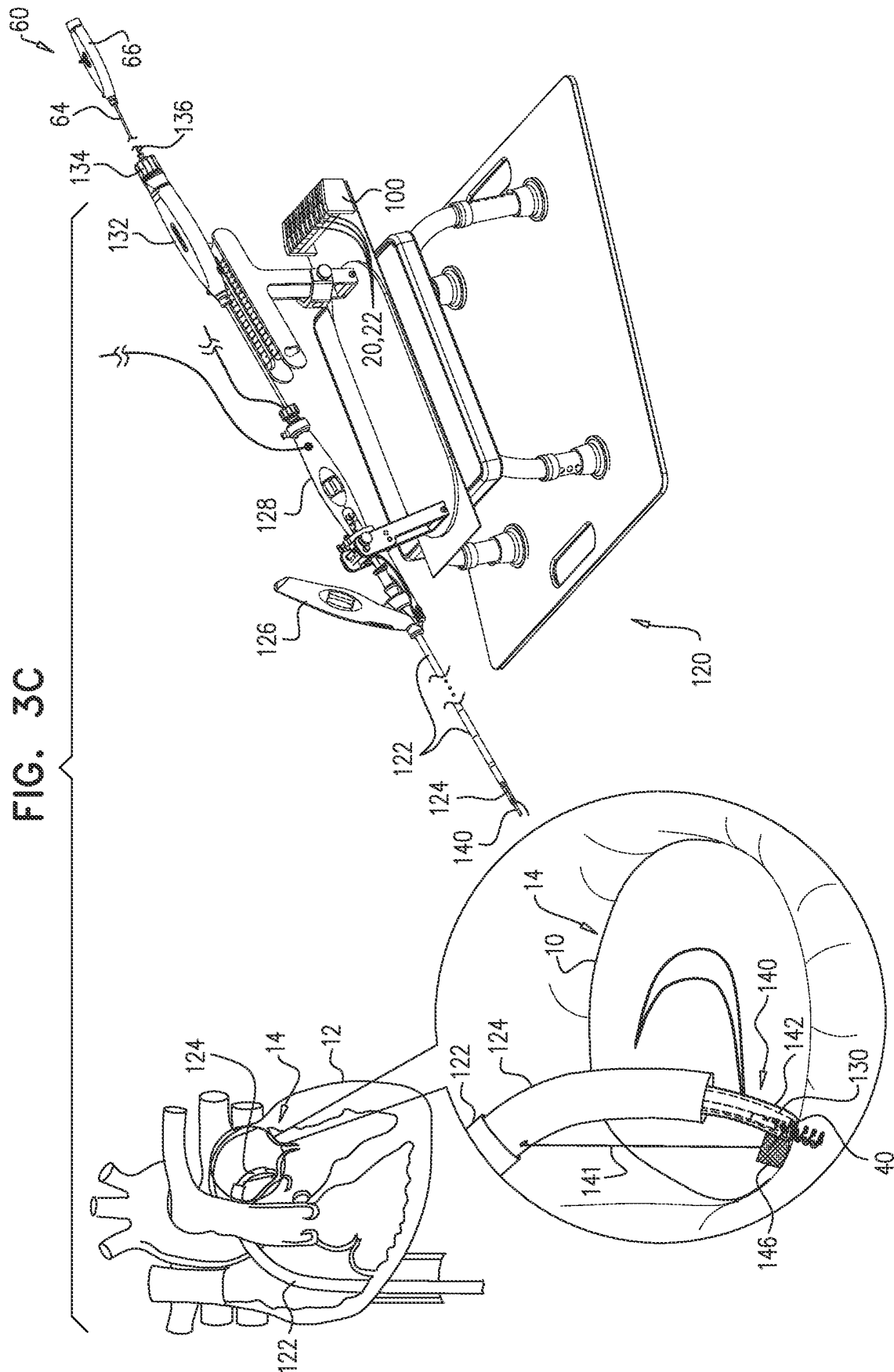
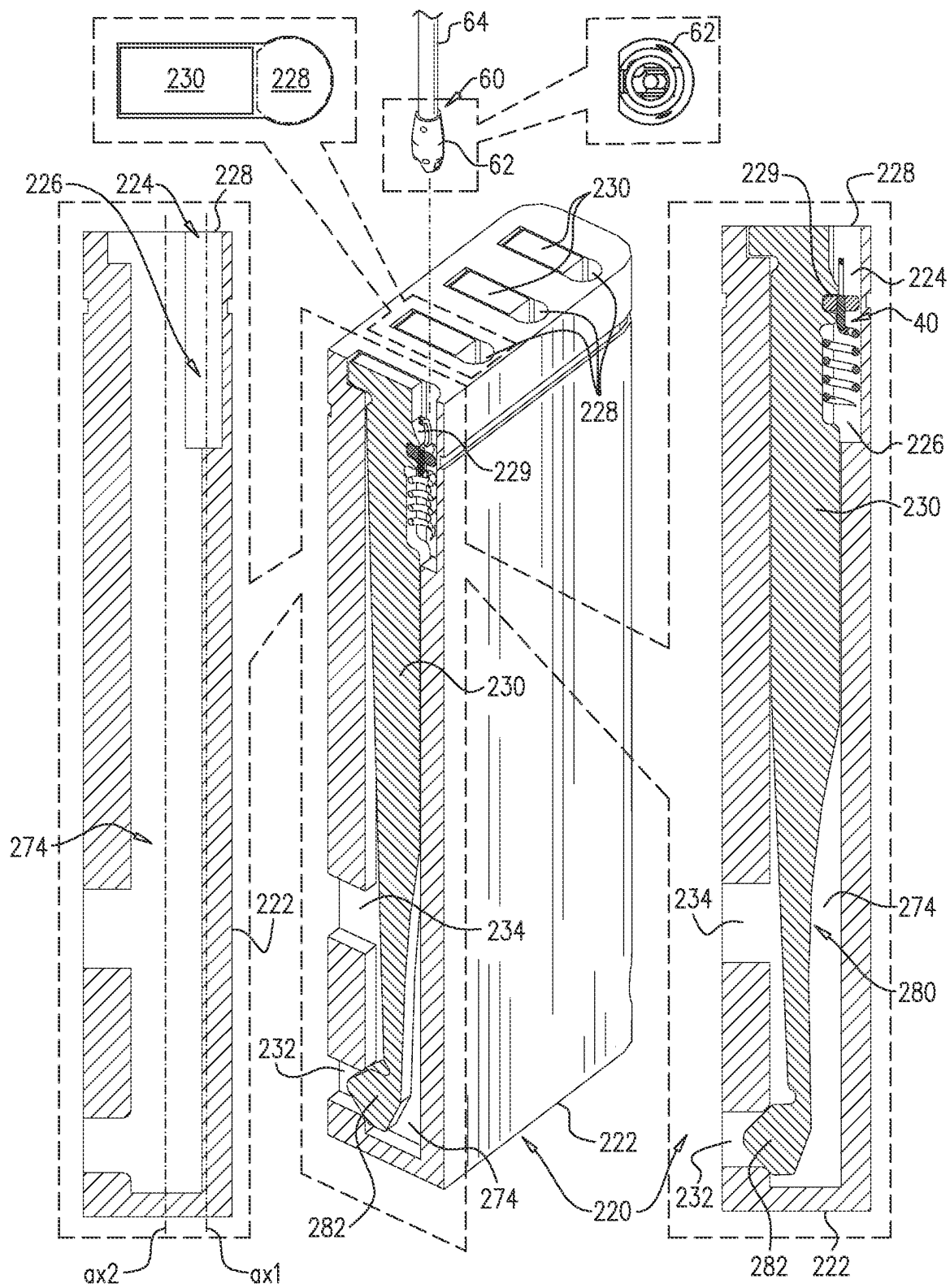


FIG. 4A



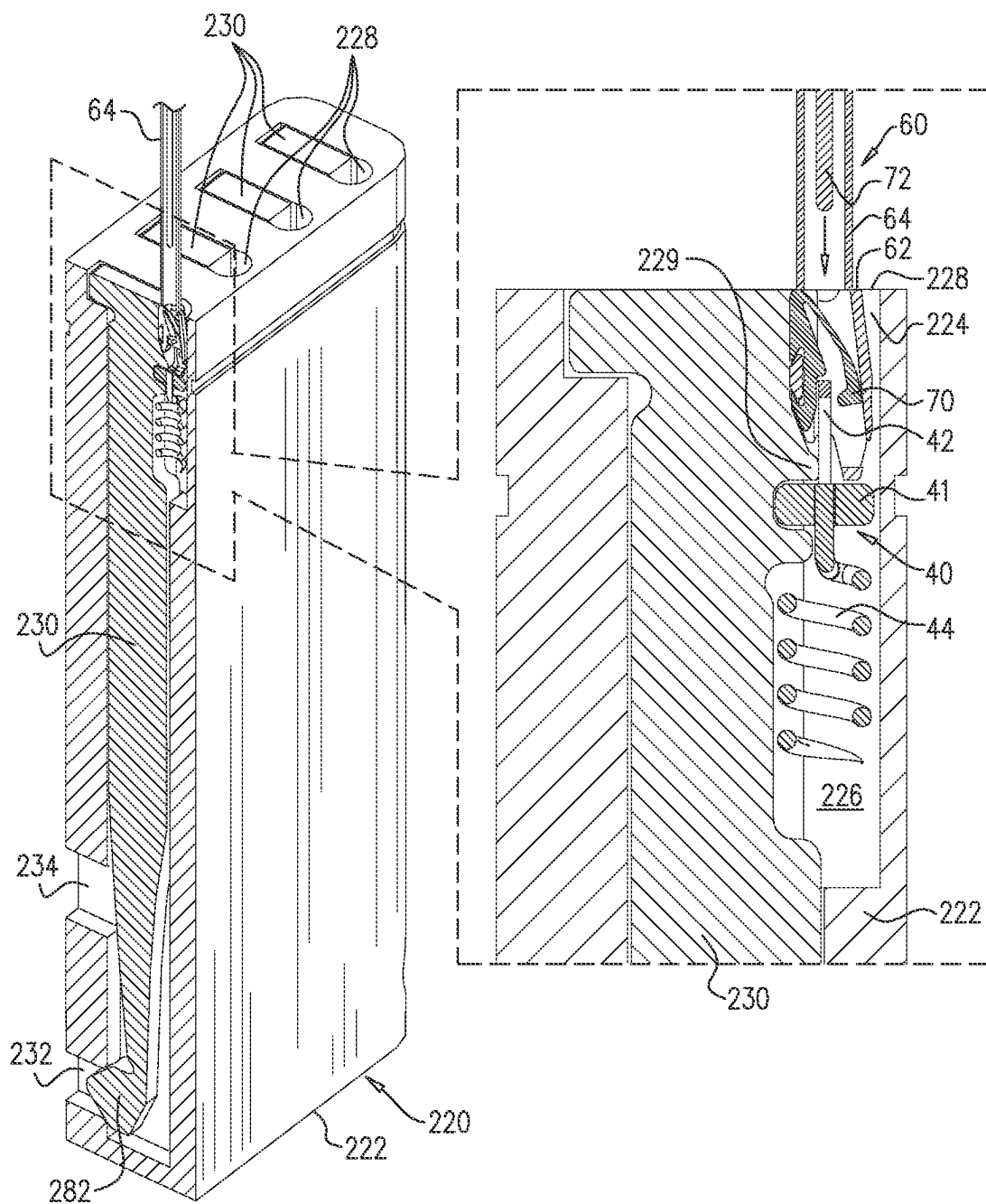


FIG. 4B

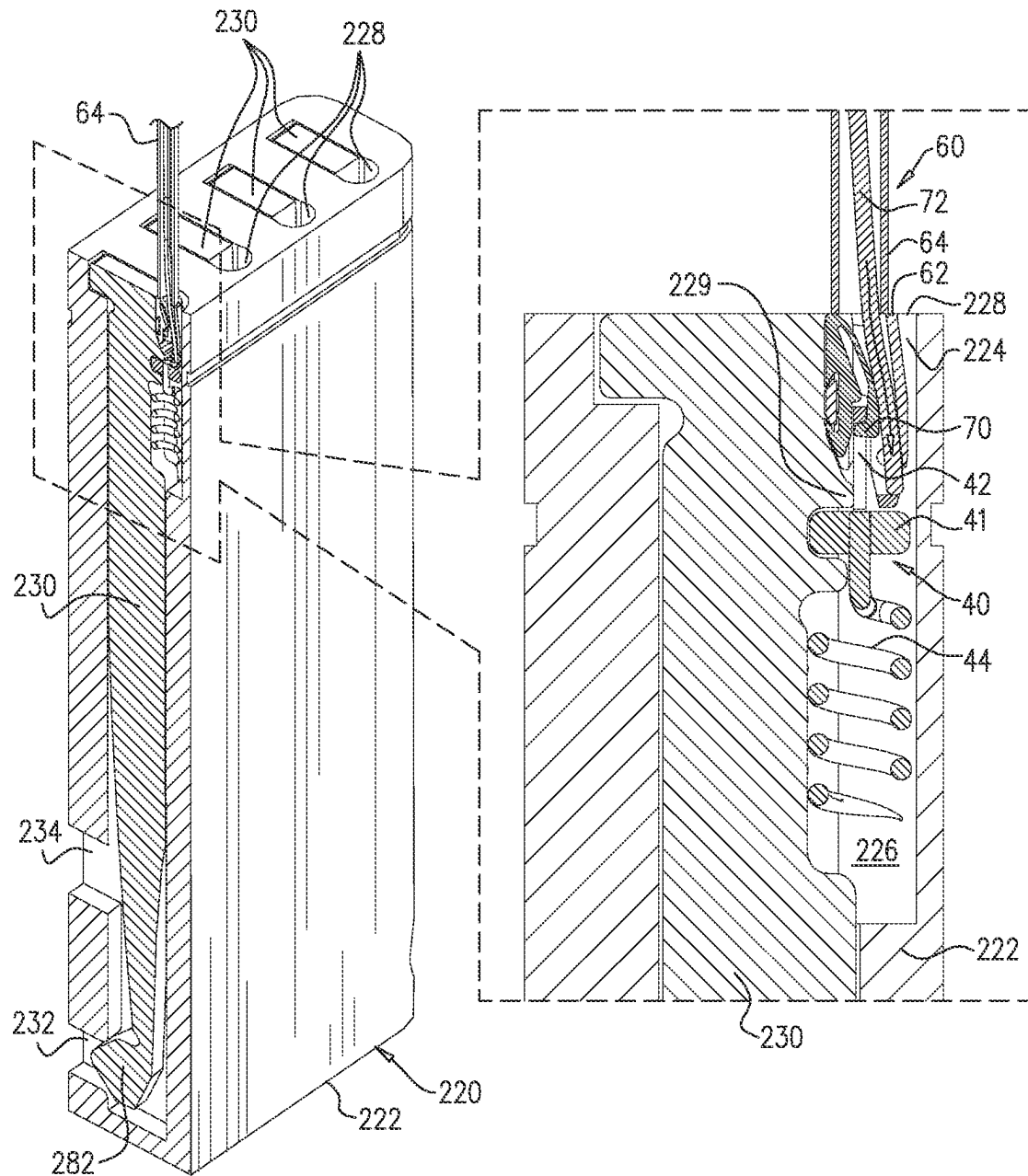


FIG. 4C

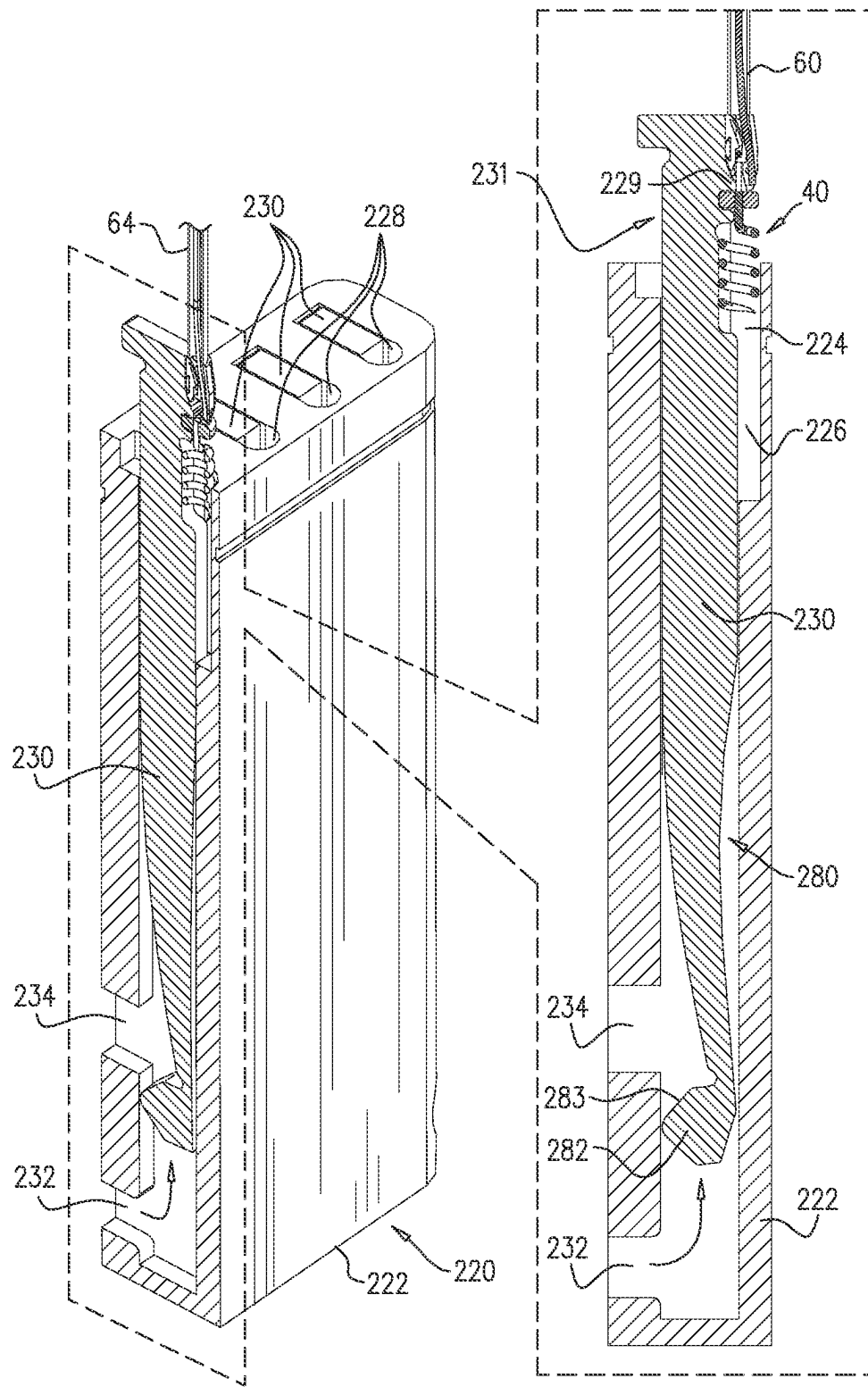
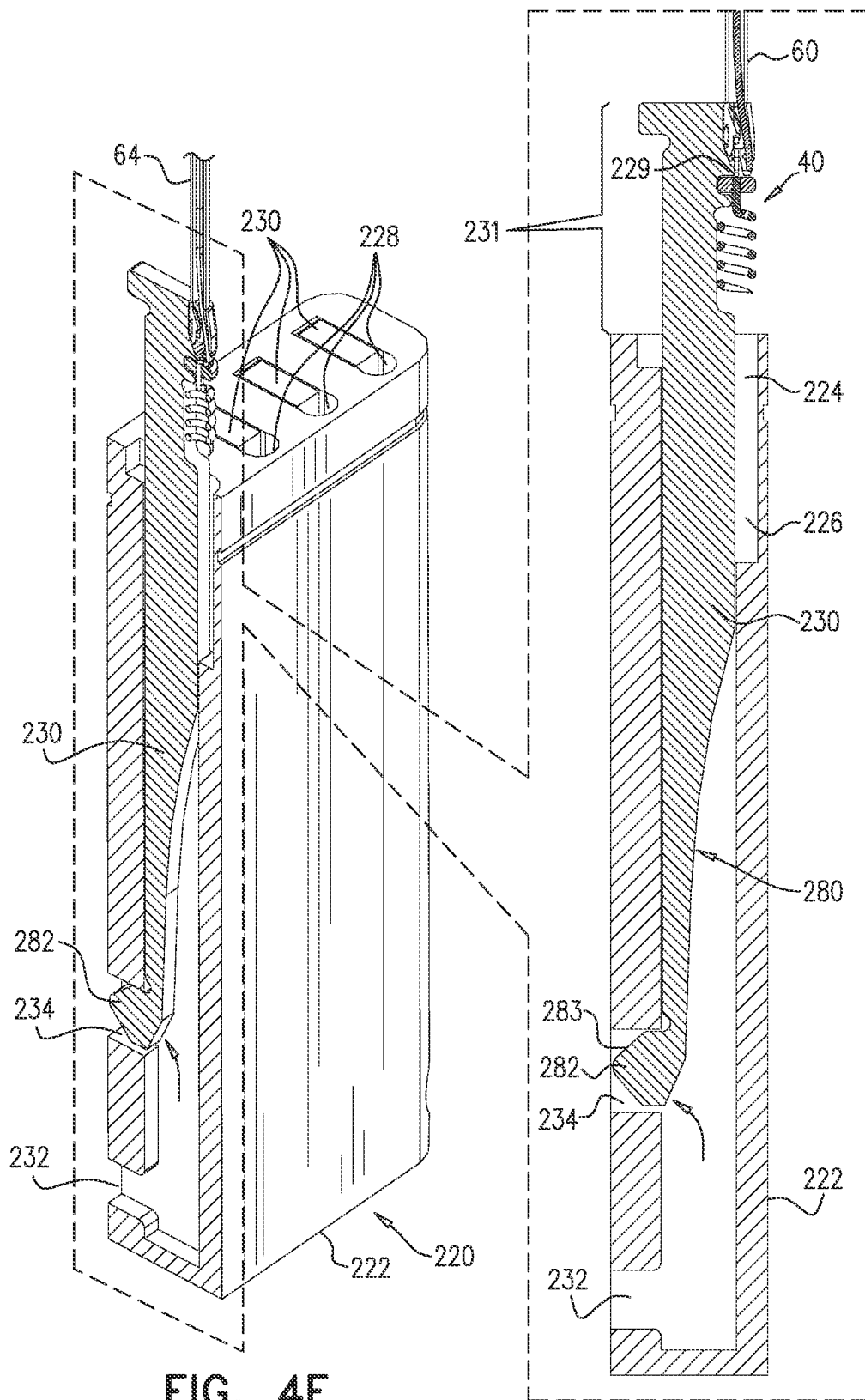
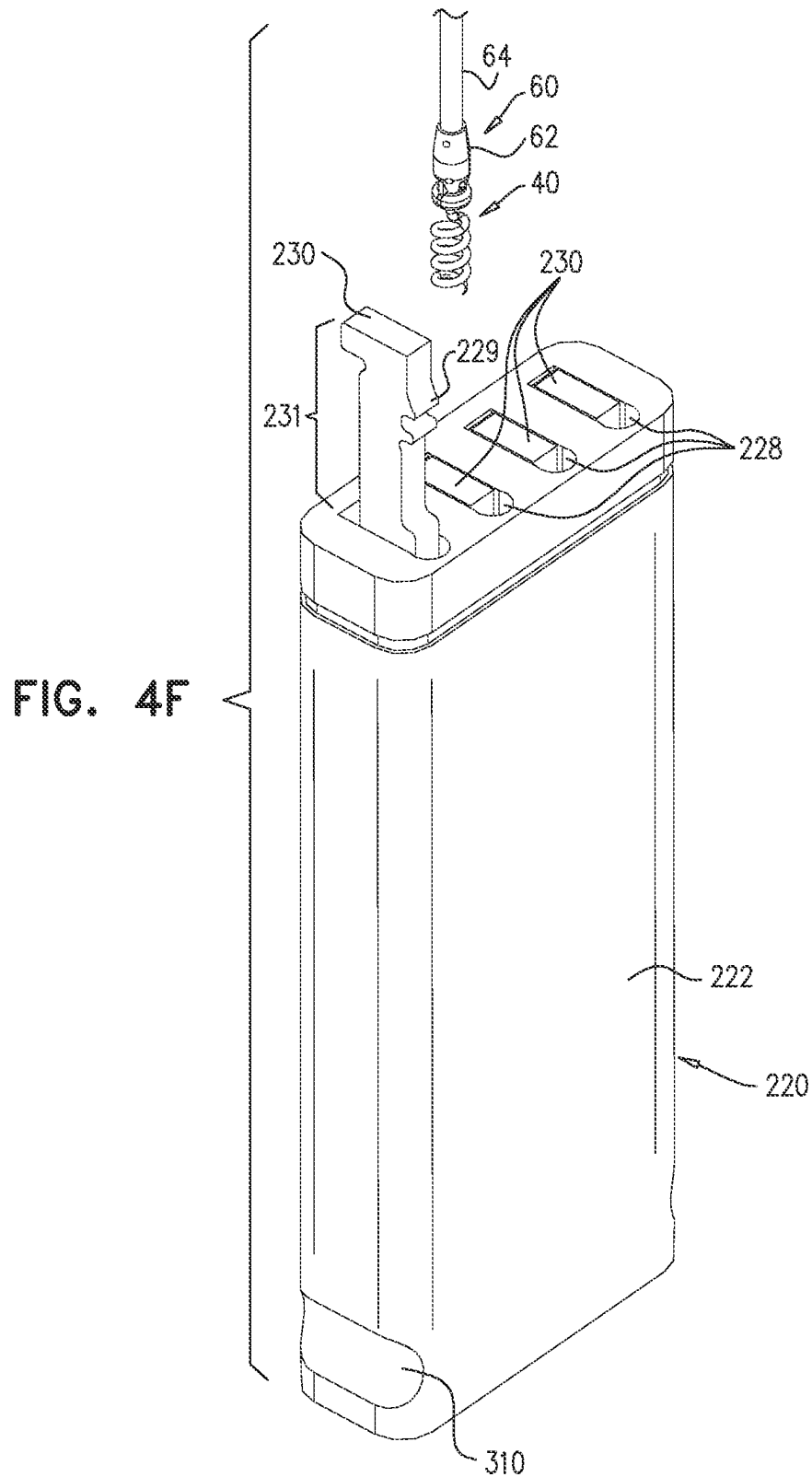


FIG. 4D





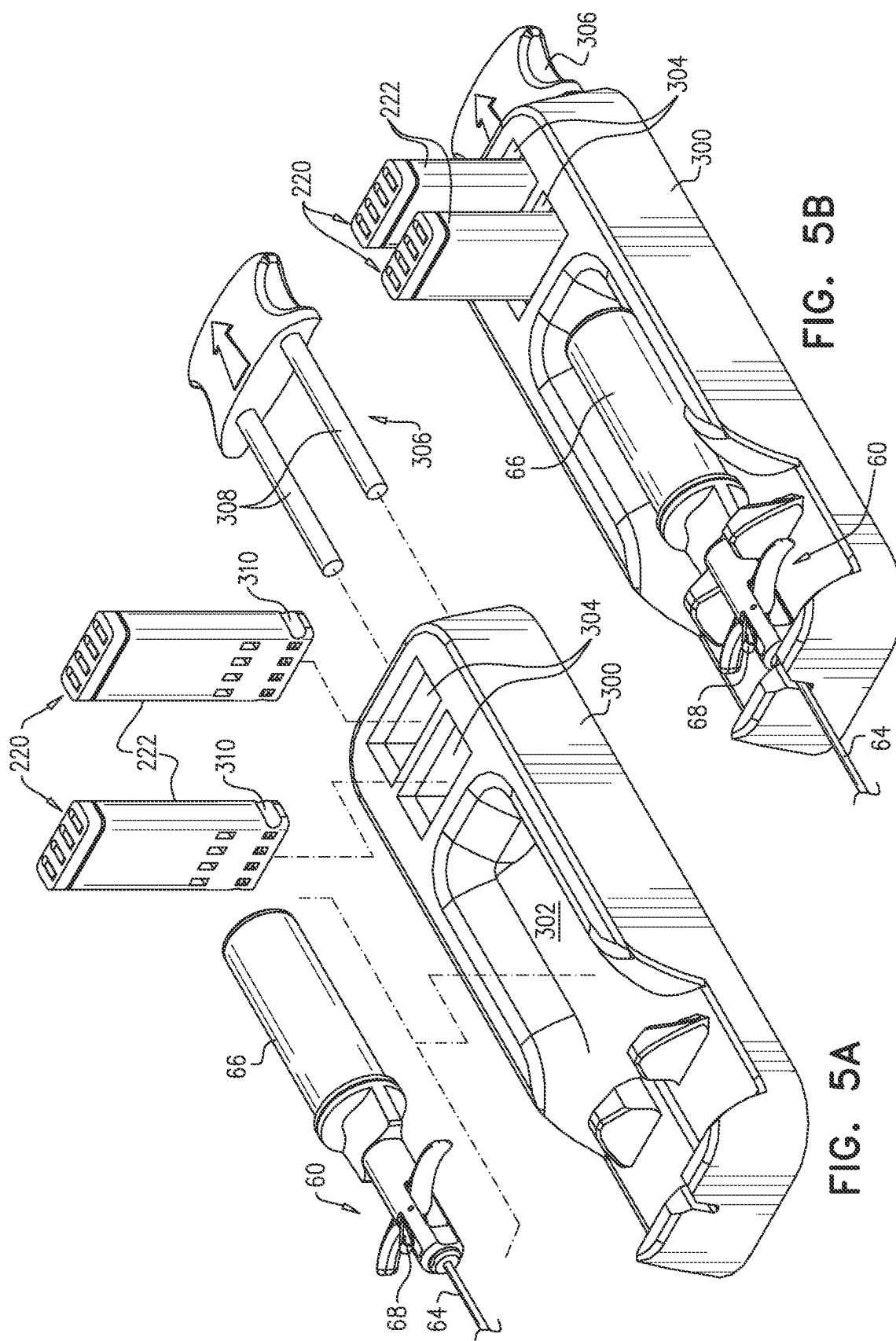
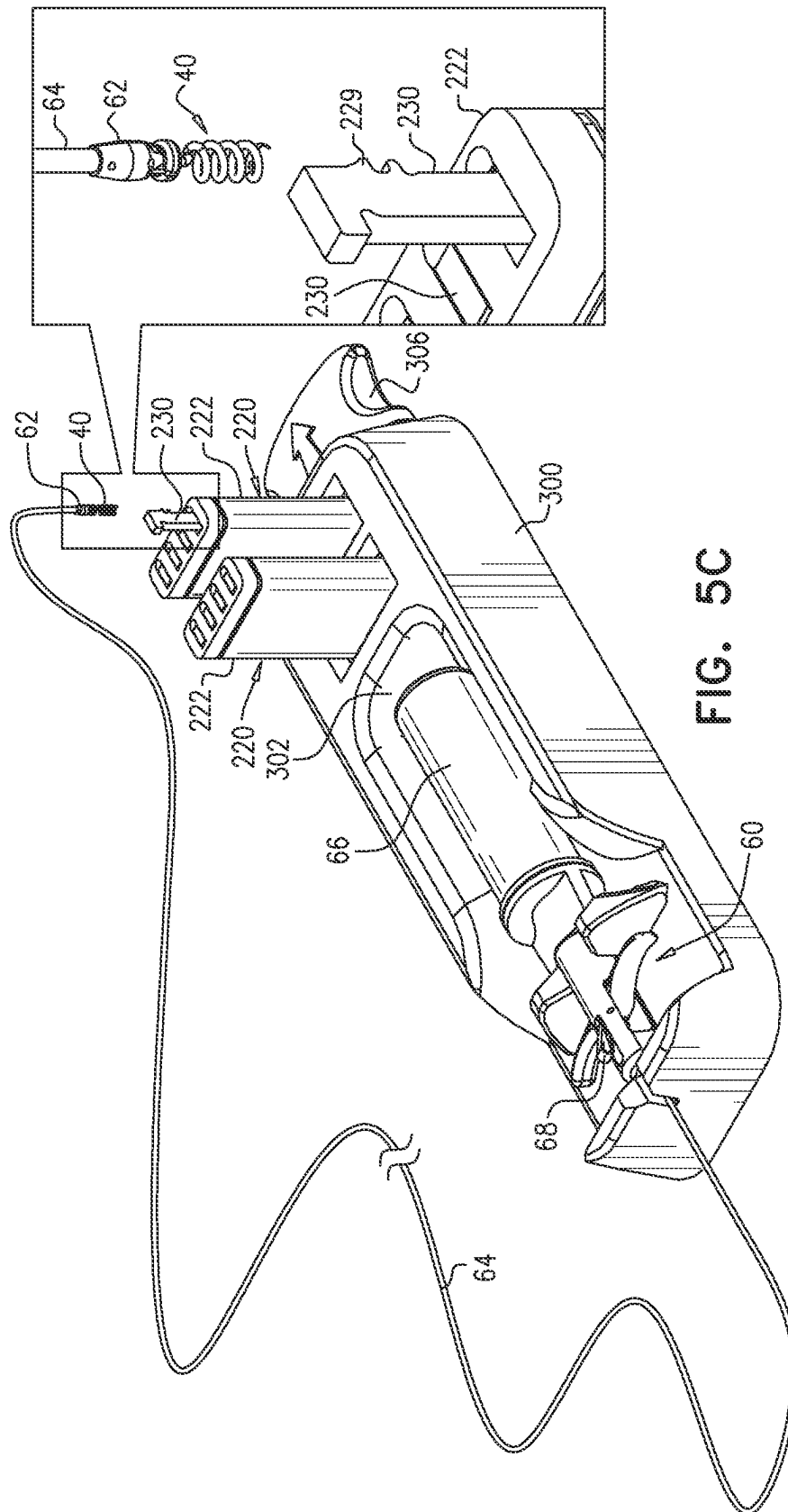


FIG. 5B

FIG. 5A



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ANCHOR MAGAZINE**CROSS-REFERENCES TO RELATED APPLICATIONS**

The present application is a Continuation of U.S. patent application Ser. No. 17/378,559 to Zipory et al., filed Jul. 16, 2021, and titled "Anchor Magazine"; which is a Continuation of U.S. patent application Ser. No. 16/239,279 to Zipory et al., filed Jan. 3, 2019, and titled "Anchor Magazine" (now U.S. Pat. No. 11,065,001); which is a Continuation of U.S. patent application Ser. No. 15/030,731 to Zipory et al., titled "Anchor Magazine" (now U.S. Pat. No. 10,299,793); which is the US National Phase of PCT application IL2014/050914 to Zipory et al., filed Apr. 20, 2016, which published as WO 2015/059699; which claims priority from US Provisional Patent Application U.S. 61/894,486 to Zipory et al., titled "Anchor Magazine", filed Oct. 23, 2013. Each of the above applications is incorporated herein by reference.

The present application is related to PCT Patent Application IL2013/050861 to Herman et al., titled "Percutaneous tissue anchor techniques", filed on Oct. 23, 2013, which published as WO 2014/064695, and PCT Patent Application IL2013/050860 to Sheps et al., titled "Controlled steering functionality for implant-delivery tool", filed on Oct. 23, 2013, which published as WO 2014/064694, both of which are incorporated herein by reference.

FIELD OF THE INVENTION

The present invention relates in general to handling of tissue anchors. More specifically, the present invention relates to devices and techniques for handling of a plurality of tissue anchors, and the use thereof at a heart valve of a patient.

BACKGROUND

Tissue anchors are placed intracorporeally so as to anchor implants to a tissue of a subject. Typically, this intracorporeal placement necessitates that the tissue anchors are small, e.g., having a greatest dimension (e.g., a length) of less than 11 mm and/or a maximum width of 3 mm. It is therefore typically advantageous to provide devices and techniques to facilitate handling of the tissue anchors.

SUMMARY OF THE INVENTION

An anchor-handling device is configured to facilitate handling of one or more tissue anchors. The anchor-handling device retains the anchors within an anchor-storage zone of a channel defined by a housing until a tool such as an anchor driver is used to retrieve the anchor. The tool is advanced through the channel, coupled to the anchor, and removed proximally out of the channel with the anchor. The anchor-handling device is configured to release (e.g., dispense) the anchor only when a proximally-directed force applied by the tool to the anchor is greater than a pre-defined threshold force (i.e., is sufficient), so as to prevent inadvertent exit of the anchor.

A retaining member is configured to retain the tissue anchor in the anchor-storage zone, typically by obstructing exit of the tissue anchor. The sufficient proximally-directed force moves the retaining member out of the way of the anchor, e.g., by moving the anchor to push the retaining member out of the way. Typically, an inhibitor inhibits movement of the retaining member, thereby configuring the

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retaining member to move out of the way of the anchor only in response to the sufficient proximally-directed force.

For some applications, the anchor-handling device is used in combination with a multi-component tubular system for transcatheter delivery of an implant, e.g., to facilitate sequential delivery of a plurality of anchors to the implant via the system.

There is therefore provided, in accordance with an application of the present invention, apparatus for use with an anchor driver, the apparatus including:

a housing, shaped to define a channel having an anchor-storage zone and a proximal opening configured to provide access for the anchor driver to the anchor-storage zone;

a tissue anchor, stored in the anchor-storage zone and slidable through the channel; and

a retaining member:

having a retaining state in which the retaining member is configured to retain the tissue anchor in the anchor-storage zone, and

being configured, by moving in response to a proximally-directed force applied to the tissue anchor, to allow the tissue anchor to leave the anchor-storage zone in response to the proximally-directed force, the proximally-directed force being greater than a pre-determined threshold force.

In an application, the apparatus is configured such that after removal of the tissue anchor from the housing, a distally-directed force required to return the apparatus to the retaining state is more than twice as great as the threshold force.

In an application, the retaining member is configured such that the threshold force is 300-1500 grams force.

In an application, the tissue anchor has a mass, and the retaining member is configured such that the threshold force, measured in grams force, is 1000-150,000 times greater than the mass of the tissue anchor, measured in grams.

In an application, the apparatus further includes an inhibitor, configured to configure the retaining member to (i) retain the tissue anchor in the anchor-storage zone, and (ii) to allow the tissue anchor to leave the anchor-storage zone in response to the proximally-directed force.

In an application, the tissue anchor is dimensioned to fit snugly in the anchor-storage zone.

In an application, the apparatus further includes a multi-component tubular system for transcatheter implantation of an implant into a subject, the implant configured to be anchored to tissue of the subject using the tissue anchor, and the housing being coupled to a component of the multi-component tubular system.

In an application, the component of the multi-component tubular system includes a stand, and the housing is coupled to the stand.

In an application, the multi-component tubular system defines a proximal port through which the anchor is introducible, and the housing is coupled to the component of the multi-component tubular system such that the proximal opening of the housing is disposed between 1 and 40 cm from the port of the multi-component tubular system.

In an application, the multi-component tubular system defines a proximal port through which the anchor is introducible, and the housing is coupled to the component of the multi-component tubular system such that the proximal opening of the housing faces generally the same direction as the port of the multi-component tubular system.

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In an application:

the housing is configured to define a plurality of channels, each of the plurality of channels having a respective anchor-storage zone and a respective proximal opening, and

the apparatus includes a plurality of tissue anchors, slidable through a respective channel and configured to be stored in a respective anchor-storage zone.

In an application, the apparatus includes a plurality of retaining members, each retaining member configured to retain a respective tissue anchor in the respective anchor-storage zone, and to allow the respective tissue anchor to leave the respective anchor-storage zone in response to a proximally-directed force applied to the respective tissue anchor.

In an application, in the retaining state, at least a portion of the retaining member obstructs proximal movement of the tissue anchor by being disposed within the channel.

In an application, the apparatus further includes the anchor driver, and in the retaining state, the anchor driver is slidable through the channel and lockable to the tissue anchor while at least the portion of the retaining member obstructs proximal movement of the tissue anchor by being disposed within the channel.

In an application, in the retaining state, the anchor driver is slidable through the channel such that a part of the anchor driver becomes positioned between a part of the tissue anchor and a part of the retaining member, and the anchor driver is lockable to the tissue anchor only while the part of the anchor driver is positioned between the part of the tissue anchor and the part of the retaining member.

In an application:

the tissue anchor includes a core, a tissue-engaging member coupled to a distal side of the core, and a coupling head coupled to a proximal side of the core, and

in the retaining state, at least the portion of the retaining member that obstructs the proximal movement of the tissue anchor obstructs the proximal movement of the tissue anchor by engaging the core.

In an application:

the housing is shaped to define a chamber that is in fluid communication with the channel,

at least part of the retaining member is configured to slide within the chamber in response to the proximally-directed force applied to the tissue anchor.

In an application, a first end of the chamber is in fluid communication with the channel, the housing defines a chamber opening of the chamber at a second end of the chamber, the portion of the retaining member includes a first portion of the retaining member, and the retaining member is configured such that, in response to the proximally-directed force applied to the tissue anchor, a second portion of the retaining member moves out of the chamber opening.

In an application, the retaining member is configured such that, in response to the proximally-directed force applied to the tissue anchor, a second portion of the retaining member moves out of the housing.

In an application, the apparatus is configured such that after removal of the tissue anchor from the housing, a distally-directed force required to return the apparatus to the retaining state is more than twice as great as the threshold force.

In an application, the retaining member includes a pin, configured to slide through the chamber.

In an application:

the housing is shaped to define a cavity that is in fluid communication with the chamber,

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at least a portion of the retaining member is resilient, the retaining member is shaped to define a detent, in the retaining state, the resilience of at least the portion of the retaining member holds the detent within the cavity, and

the retaining member is configured to deform in response to the proximally-directed force applied to the tissue anchor, such that the detent exits the cavity.

In an application:

the cavity includes a first cavity,

the housing is shaped to define a second cavity that is in fluid communication with the chamber, and

the apparatus is dimensioned such that when the retaining member allows the tissue anchor to leave the anchor-storage zone, further proximal movement of the retaining member causes the detent to move into the second cavity.

In an application, the second cavity is larger in at least one dimension than the first cavity.

In an application, the second cavity is differently shaped to the first cavity.

In an application, the second cavity and the detent are dimensioned such that when the detent is disposed within the second cavity, a distally-directed force required to return the apparatus to the retaining state is more than twice as great as the threshold force.

In an application, at least a portion of the pin is dimensioned to slide snugly through the chamber.

In an application, the apparatus further includes an inhibitor tongue having a pin-contacting portion that is in contact with the pin, and configured to (i) inhibit the pin from sliding through the chamber in response to a sub-threshold force, and (ii) to allow the pin to slide through the chamber in response to the proximally-directed force applied to the tissue anchor.

In an application:

the pin is shaped to define a cavity,

at least a portion of the inhibitor tongue is resilient, in the retaining state, the resilience of at least the portion of the inhibitor tongue holds the pin-contacting portion within the cavity, and

the inhibitor tongue is configured to deform in response to the proximally-directed force applied to the tissue anchor, such that the pin-contacting portion exits the cavity.

In an application:

the cavity includes a first cavity,

the pin is shaped to define a second cavity,

the apparatus is dimensioned such that when the retaining member allows the tissue anchor to leave the anchor-storage zone, further proximal movement of the retaining member causes the pin-contacting portion to move into the second cavity.

In an application, the second cavity is larger in at least one dimension than the first cavity.

In an application, the second cavity is differently shaped to the first cavity.

In an application, the second cavity and the pin-contacting portion are dimensioned such that when the pin-contacting portion is disposed within the second cavity, a distally-directed force required to return the apparatus to the retaining state is more than twice as great as the threshold force.

In an application, the chamber is in fluid communication with the channel at a distal end of the chamber, and has a proximal-distal longitudinal axis that is disposed at between 5 and 30 degrees from a proximal-distal longitudinal axis of the channel.

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In an application, the proximal-distal longitudinal axis of the chamber is disposed at between 5 and 20 degrees from the proximal-distal longitudinal axis of the channel.

In an application, the proximal-distal longitudinal axis of the chamber is disposed at between 11 and 14 degrees from the proximal-distal longitudinal axis of the channel.

In an application, a central longitudinal axis of the chamber is parallel with a central longitudinal axis of the channel.

In an application, the tissue anchor is dimensioned to fit snugly through the channel.

In an application, the tissue anchor includes a core, a tissue-engaging member coupled to a distal side of the core, and a coupling head, the core is dimensioned to fit snugly through the channel, and the tissue-engaging member is dimensioned so as to not touch the housing when the tissue anchor moves through the channel.

In an application, the apparatus further includes the anchor driver.

In an application, the anchor driver includes:

at a distal end thereof, an anchor-engaging head introducible through the opening of the housing and actuatable to be reversibly coupled to the tissue anchor;

at a proximal end thereof, a handle including an adjuster configured to actuate the anchor-engaging head; and a flexible shaft;

disposed between the distal end of the anchor driver and the proximal end of the anchor driver,

having a length of 50-250 cm, and

configured to be transcatheterally advanced through vasculature of a subject.

In an application, the opening of the housing is rotationally asymmetrical, a transverse cross-section of the anchor-engaging head is rotationally asymmetrical, and the opening limits a range of rotational orientations of the anchor-engaging head with respect to the opening in which the anchor-engaging head is introducible through the opening.

In an application, the opening of the housing and the transverse cross-section of the anchor-engaging head each have the shape of an ellipse that has had a segment removed.

In an application, the tissue anchor is stored in the anchor-storage zone in a given rotational orientation of the tissue anchor with respect to the opening, the anchor-engaging head is couplable to the tissue anchor in not all rotational orientations of the head with respect to the tissue anchor, and the anchor-engaging head is couplable to the tissue anchor without rotating the anchor-engaging head subsequently to introducing the anchor-engaging head through the opening.

In an application, the opening limits the range of rotational orientations such that the anchor-engaging head is introducible through the opening in only a given rotational orientation of the head with respect to the opening.

In an application, the apparatus further includes a base, and:

the housing is couplable to the base,

the base is configured to at least partly immobilize the housing, and

the base is shaped to define a receptacle for housing and at least partly immobilizing the handle.

In an application, when the housing is coupled to the base, the housing is disposed less than 30 cm from the receptacle.

In an application:

the receptacle is a handle receptacle,

the housing is reversibly couplable to the base,

the base is shaped to further define a housing receptacle, configured to house the housing,

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the base further includes a locking element, movable between a locked state that locks the housing within the receptacle, and an unlocked state that facilitates release of the housing from the receptacle.

In an application, the adjuster is operable while the receptacle houses the handle.

In an application, the apparatus is configured such that while the receptacle houses the handle, a human operator may:

with a first hand of the operator, grasp a distal portion of the driver and introduce the head into the opening, and with a second hand of the operator, reversibly actuate the head by operating the adjuster while grasping the distal portion of the driver with the first hand.

There is further provided, in accordance with an application of the present invention, apparatus for use with an anchor driver, the apparatus including:

a housing, shaped to define a channel having (a) an anchor-storage zone and (b) a proximal opening configured to provide access for the anchor driver to the anchor-storage zone;

a tissue anchor, slidable through the channel and configured to be stored in the anchor-storage zone; and

a retaining member:

having a retaining state in which the retaining member is configured to retain the tissue anchor in the anchor-storage zone, and

being disposed within the housing such that sliding of the tissue anchor proximally out of the anchor-storage zone and through the channel causes the retaining member to slide in an at least partly proximal direction.

In an application, the retaining member is disposed within the housing such that sliding of the tissue anchor proximally out of the anchor-storage zone and through the channel causes the retaining member to slide along an axis that is disposed at an angle of less than 30 degrees with respect to a central longitudinal axis of the channel.

There is further provided, in accordance with an application of the present invention, apparatus, including:

a housing, shaped to define a channel having an anchor-storage zone and a proximal opening;

an anchor driver including an anchor-engaging head, a handle, and a shaft therebetween, and:

the shaft is flexible and is configured to be transluminally advanced into a subject, and

the anchor-engaging head is dimensioned to be advanceable through the proximal opening toward the anchor-storage zone; and

a tissue anchor:

stored in the anchor-storage zone,

including (i) a coupling head configured to be locked to the anchor-engaging head while the tissue anchor is in the anchor-storage zone, and (ii) a tissue-engaging member configured to be driven into tissue of the subject using the anchor driver, and

configured such that, while stored in the anchor-storage zone, the tissue anchor is movable out of the anchor-storage zone toward the proximal opening only in response to a proximally-directed force being applied to the tissue anchor, the proximally-directed force being greater than a pre-determined threshold force.

In an application, the tissue anchor is configured to be movable out of the anchor-storage zone only in response to the proximally-directed force, by being dimensioned with respect to at least one dimension of the housing such that the

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tissue anchor is movable out of the anchor-storage zone only in response to the proximally-directed force.

There is further provided, in accordance with an application of the present invention, apparatus, including:

an anchor-handling device including a housing, shaped to define a channel having an anchor-storage zone and a proximal opening;

an anchor driver including an anchor-engaging head, a handle, and a shaft therebetween, and:

the shaft is flexible and is configured to be transluminally advanced into a subject, and

the anchor-engaging head is dimensioned to be advanceable through the proximal opening toward the anchor-storage zone; and

a tissue anchor:

stored in the anchor-storage zone,

including (i) a coupling head configured to be locked to the anchor-engaging head while the tissue anchor is in the anchor-storage zone, and (ii) a tissue-engaging member configured to be driven into tissue of the subject using the anchor driver, and

configured such that, while stored in the anchor-storage zone, the tissue anchor is movable out of the anchor-storage zone toward the proximal opening in response to

a proximally-directed force being applied to the tissue anchor by the anchor driver, and the anchor-handling device is configured to provide an indication of the movement of the tissue anchor out of the anchor-storage zone toward the proximal opening in response to the proximally-directed force.

In an application, the anchor-handling device is configured to provide the indication by the housing being at least in part transparent, such that the movement of the tissue anchor is viewable from outside the housing.

In an application, the anchor-handling device is configured to provide the indication by including an element that moves with respect to the housing in response to the movement of the tissue anchor.

In an application, the element that moves with respect to the housing moves out of the housing in response to the movement of the tissue anchor.

In an application, the element that moves with respect to the housing moves with respect to the housing at a rate that is relative to a rate at which the anchor moves with respect to the housing.

The present invention will be more fully understood from the following detailed description of applications thereof, taken together with the drawings, in which:

BRIEF DESCRIPTION OF THE DRAWINGS

FIGS. 1A-F are schematic illustrations of an anchor-handling device, configured to facilitate handling of at least one tissue anchor, in accordance with some applications of the invention;

FIG. 2 is a schematic illustration of a multiple-anchor-handling device, in accordance with some applications of the invention;

FIGS. 3A-C are schematic illustrations of a system for transcatheter delivery of an implant, and anchoring of the implant using an anchor driver and a plurality of anchors provided in the multiple-anchor-handling device of FIG. 2, in accordance with some applications of the invention;

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FIGS. 4A-F are schematic illustrations of an anchor-handling device, configured to facilitate handling of at least one tissue anchor, in accordance with some applications of the invention; and

FIGS. 5A-C are schematic illustrations of a base to which an anchor-handling device is couplable, and which is configured to at least partly immobilize the anchor-handling device, in accordance with some applications of the invention.

DETAILED DESCRIPTION OF EMBODIMENTS

Reference is made to FIGS. 1A-F, which are schematic illustrations of an anchor-handling device 20, configured to facilitate handling of at least one tissue anchor 40, in accordance with some applications of the invention. Device 20 comprises a housing 22 that defines a channel 24, an anchor-storage zone 26 (e.g., at a distal end of the channel) and an opening 28 (e.g., at a proximal end of the channel) that provides access to the channel and the anchor-storage zone. Typically, there is a smooth transition between anchor-storage zone 26 and channel 24. Device 20 further comprises a retaining member, such as a pin 30, which is configured to retain tissue anchor 40 in zone 26, and to stop retaining the tissue anchor in response to a sufficient proximally-directed force applied to the tissue anchor. That is, when a proximally-directed force that is greater than a pre-determined threshold force is applied to tissue anchor 40, the retaining member stops retaining (e.g., releases) the tissue anchor.

Typically, the retaining member (e.g., pin 30) has a retaining state in which it retains tissue anchor 40 within zone 26, and is moved out the retaining state when the sufficient proximally-directed force is applied to the tissue anchor. FIG. 1A shows, in accordance with some applications of the invention, pin 30 in a retaining state thereof, in which at least a portion 29 (e.g., an obstructing portion, and/or a distal portion) of the pin is disposed within channel 24 (e.g., proximal to anchor 40), thereby retaining the anchor in zone 26 by obstructing proximal movement of the tissue anchor. Typically, portion 29 obstructs proximal movement of anchor 40 by engaging and/or obstructing core 41 of the anchor (described hereinbelow).

It is to be noted that although pin 30 is shown as being generally cylindrical (i.e., having a generally circular transverse cross-section), the term "pin", as used throughout the present application, including the specification and the claims, may include a pin having a different shape (e.g., having a noncircular transverse cross-section). For example, pin 230 (described hereinbelow with reference to FIGS. 4A-5C) typically has a rectangular cross-section.

FIG. 1A shows an anchor driver 60 being advanced toward opening 28 of channel 24, and FIG. 1B shows the anchor driver having been further advanced into and through channel 24, to anchor 40. As more clearly shown in FIGS. 1A and 1F, opening 28 is typically beveled (i.e., disposed at an angle smaller than 90 degrees to a longitudinal axis of channel 24), such that the opening has a greater area than does a transverse cross section of the channel, thereby facilitating introduction of driver 60 into the channel. It is to be noted that the shape of opening 28 provides a proximal region 25 in which channel 24 is at least half open on a lateral side (i.e., at least half of the circumferential surface of the channel is missing). This shape thereby facilitates placement of a driver head 62 of driver 60 in proximal region 25 of channel 24, e.g., by reducing a requirement for the driver to be aligned with the channel before its intro-

duction into the channel. Driver 60 (e.g., head 62 thereof) can subsequently be advanced further into channel 24, using region 25 as a guide track.

Driver 60 typically comprises an anchor-engaging head 62 at a distal end of the driver, and a shaft 64 proximal to the anchor-engaging head. Shaft 64 is flexible and advanceable (e.g., transcatheterally) through vasculature of a subject, and typically has a length greater than 20 cm, and/or less than 2.5 m, such as greater than 50 cm and/or less than 1.5 m, e.g., between 0.9 m and 1.2 m. For some applications, driver 60 comprises a handle 66 at a proximal end of shaft 64, the handle comprising an adjuster 68 (e.g., a switch or a lever) configured to actuate engaging head 62.

Tissue anchor 40 typically comprises a core 41, a tissue-engaging member 44 coupled to a distal side of the core, and a coupling head 42 coupled to a proximal side of the core. Engaging head 62 is configured to be reversibly couplable to tissue anchor 40 (e.g., to coupling head 42 thereof), so as to facilitate acquisition of the anchor from device 20, driving of the anchor into tissue of the subject, and subsequent release of the anchor and withdrawal of driver 60 from the subject. For example, actuation of engaging head 62 by adjuster 68 may comprise transitioning the engaging head between (i) an open state in which the engaging head is configured to receive and/or release anchor 40 (FIGS. 1A-B), and (ii) a closed state in which the engaging head, having received the anchor (FIG. 1C), is coupled (e.g., locked) to the anchor.

FIG. 1B shows engaging head 62 having received anchor 40, but not yet coupled (e.g., locked) to the anchor. It is to be noted that in this position part of driver 60 (e.g., head 62) is disposed (e.g., sandwiched) between part of the retaining member (e.g., portion 29) and part of the anchor (e.g., coupling head 42).

FIG. 1C shows engaging head 62 being coupled (e.g., locked) to anchor 40. For some applications of the invention, engaging head 62 comprises a detent 70 that is transitioned into the closed state when a controller, such as a rod or a wire 72, is moved distally by adjuster 68, and automatically transitions back into the open state when the wire is withdrawn. FIG. 1C shows wire 72 having been moved distally into engaging head 62, and detent 70 having been pushed into the closed state, thereby coupling the engaging head to anchor 40 (e.g., to coupling head 42 thereof).

FIGS. 1D-E show anchor 40 being withdrawn proximally from zone 26 of channel 24 by the sufficient proximally-directed force being applied to the anchor by driver 60. As described hereinabove, in response to the sufficient proximally-directed force applied to the tissue anchor (i.e., if the proximally-directed force is greater than the pre-determined threshold force), the retaining member (e.g., pin 30) stops retaining the tissue anchor in zone 26. For example, and as shown in FIGS. 1D-E, the sufficient proximally-directed force overcomes the retention provided by pin 30 and pushes portion 29 of pin 30 out of the channel and into a chamber 74 that is in fluid communication with the channel (at least part of pin 30 thereby sliding within the chamber).

It is hypothesized that this configuring of device 20 to require that the sufficient proximally-directed force be applied to tissue anchor 40 prevents inadvertent movement and/or exit of the tissue anchor (e.g., due to general transport or handling of the device), and/or withdrawal of the anchor by driver 60 when the driver is sub-optimally coupled to the anchor.

For some applications, a first end 76 of chamber 74 is in fluid communication with channel 24, housing 22 defines an opening 78 at a second end of the chamber, and the pushing

of portion 29 of pin 30 by the sufficient proximally-directed force pushes a second (e.g., proximal) portion 31 of the pin out of opening 78. This feature and advantages thereof are described in more detail hereinbelow. Typically, chamber 74 has a proximal-distal longitudinal axis that is disposed at between 5 and 30 degrees, e.g., 5-20 degrees (e.g., 5-15 degrees or 10-degrees, such as between 11 and 14 degrees) with respect to the longitudinal axis of channel 24. It is hypothesized that, for some applications, this angular disposition of the channel and chamber facilitates the above described movement of pin 30 in response to the sufficient proximally-directed force applied to the tissue anchor.

FIG. 1F shows anchor 40 having been fully withdrawn out of channel 24 via opening 28. Once anchor 40 has been fully withdrawn, driver 60 may be used to anchor tissue anchor 40 to tissue of a subject, e.g., by driving tissue-engaging member 44 (FIG. 1A) of the anchor into the tissue. It may be used as a tissue anchor as is known in the art. For example, using driver 60, anchor 40 may be advanced through a transluminal implant-delivery system and used to couple an implant to tissue of a subject, e.g., as described hereinbelow with reference to FIGS. 3A-C.

For some applications, device 20 comprises an inhibitor, configured to configure the retaining member (e.g., pin 30) to (i) retain the tissue anchor in anchor-storage zone 26, and (ii) to stop retaining the tissue anchor in response to the sufficient proximally-directed force. For example, the inhibitor may comprise an inhibitor tongue 80, that has a pin-contacting portion 82 (e.g., a pin-contacting surface) that is in contact with pin 30, and that provides resistance that (i) inhibits sliding of the pin through chamber 74 (e.g., prevents sliding of the pin in response to an insufficient proximally-directed force, i.e., a proximally-directed force that is less than the pre-determined threshold force), and (ii) allows sliding of the pin through the chamber in response to the sufficient proximally-directed force that is greater than the pre-determined threshold force being applied to tissue anchor 40. Pin-contacting portion 82 is typically held in contact with pin 30 by a spring mechanism. For example, and as shown in FIGS. 1A-F, inhibitor tongue 80 may comprise an elastically-deformable (e.g., shape-memory) material, and may be coupled to housing 22 by in a manner in which the inhibitor tongue itself provides the spring mechanism.

For some applications, pin 30 defines a cavity 32 therein (e.g., a recess or a notch in a lateral side of the pin), in which pin-contacting portion 82 is typically disposed while anchor is disposed within anchor-storage zone 26 (e.g., in a state in which the device is provided). For such applications, portion 82 serves as a detent. For such applications, cavity 32 and inhibitor tongue 80 are configured such that when a proximally-directed force equal to or greater than the threshold force is applied to anchor 40, pin 30 is pushed against pin-contacting portion 82, and inhibitor tongue 80 responsively deforms such that the pin-contacting portion moves out of cavity 32, allowing pin 30 to move further proximally (FIG. 1D). Typically, for such applications, once portion 82 has moved out of cavity 32, a proximally-directed force that is smaller than the threshold force is sufficient to move pin 30 further proximally. That is, once the initial resistance provided by the inhibitor is overcome, anchor 40 is further withdrawable using a smaller force than that required to overcome the initial resistance.

It will be understood by those skilled in the art that it is possible to use other configurations to achieve a behavior similar to that described above. For example, housing 22

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may define a cavity, and pin 30 may comprise a flexible protrusion that extends into the cavity of the housing.

For some applications, the inhibitor (e.g., tongue 80) provides the resistance by applying friction against the retaining member (e.g., pin 30). For example, pin-contacting portion 82 may comprise a high-friction pin-contacting surface.

Reference is made to FIG. 2, which is a schematic illustration of a multiple-anchor-handling device 100, in accordance with some applications of the invention. Device 100 defines a plurality of channels 24, each channel having a respective proximal opening 28 and a respective anchor-storage zone 26 that is configured to store a respective tissue anchor 40 (zone 26 and anchor 40 not visible in FIG. 2). Typically, device 100 further comprises a plurality of retaining members (e.g., pins 30), each retaining member being configured to retain a respective tissue anchor in its respective anchor-storage zone 26, and to stop retaining the respective tissue anchor in response to the sufficient proximally-directed force being applied to its respective tissue anchor. For some applications, device 100 comprises a plurality of devices 20. For example, device 100 may comprise a plurality of housings 22, each housing defining exactly one channel 24 and exactly one retaining member (e.g., pin 30).

As described hereinabove, for some applications, the sufficient proximally-directed force pushes a second (e.g., proximal) portion 31 of pin 30 out of opening 78 of chamber 74. Therefore, when driver 60 is withdrawn proximally, movement of portion 31 toward and/or out of opening 78 indicates that anchor-engaging head 62 has been successfully coupled to tissue anchor 40, and that the tissue anchor is also being withdrawn proximally. Thus, during an initial partial withdrawal of driver 60, movement of portion 31 toward and/or out of opening 78 provides an indication to the operator (e.g., physician) to continue to withdraw driver 60, whereas absence of such movement of portion 31 provides an indication to the operator to reattempt coupling of the driver to tissue anchor 40.

Following removal of anchor 40 from channel 24, portion 31 remains exposed from opening 78. This may be particularly useful for a physician using a multiple-anchor-handling device, such as device 100, e.g., to prevent the physician inadvertently attempting to obtain an anchor from an empty zone 26. That is, portion 31 functions as an empty-housing indicator.

For some applications, pin 30 defines a second cavity 34 therein (e.g., a second notch in a lateral side of the pin), disposed closer to distal portion 29 than is cavity 32. Second cavity 34 is positioned such that when (1) distal portion 29 is no longer obstructing anchor 40, and (2) second portion 31 is exposed out of opening 78, pin-engaging portion 82 of inhibitor tongue 80 moves into the second cavity (e.g., as shown in FIG. 1E). In this state, tongue 80 inhibits pin 30 from moving distally back into housing 22, thereby increasing the reliability of portion 31 functioning as an empty-housing indicator. For some applications, and as shown in FIGS. 1A-F, second cavity 34 is shaped such that once portion 82 has moved into (e.g., engaged) second cavity 34, the moving of portion 31 back into housing 22 requires a distally-directed force that is more than twice as great (in the opposite direction) as the threshold force that was previously required to move portion 31 out of the housing. For example, the moving of portion 31 back into housing 22 may be in effect prevented.

Reference is made to FIGS. 3A-C, which are schematic illustrations of a multi-component tubular system 120 for

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transcatheter delivery of an implant 140, and anchoring of the implant using anchor driver 60 and a plurality of anchors 40 provided in multiple-anchor-handling device 100, in accordance with some applications of the invention. Implant 140 comprises an annuloplasty structure comprising a sleeve 142, a flexible elongated contracting member 144 that extends along the sleeve, and an adjustment mechanism 146 which facilitates contracting and expanding of the annuloplasty structure. Typically, adjustment mechanism 146 comprises a spool around which successive portions of member 144 are wound in order to contract the annuloplasty structure after implantation.

Implant 140 is configured to be anchored to an annulus 10 of a valve of the heart 12 of a subject, such as a mitral valve 14 of the subject, and to change a dimension of the annulus when contracted or expanded using adjustment mechanism 146.

System 120 comprises one or more steerable catheters, and typically comprises an outer catheter 122 and an inner catheter 124 that is advanceable through the outer catheter. Outer catheter 122 is advanceable and steerable using a first handle 126, and inner catheter 124 is advanceable and steerable using a second handle 128. Typically, second handle 128 is couplable (e.g., lockable) to first handle 126, e.g., after advancement of catheter 124 through catheter 122.

Implant 140 is typically (1) advanceable through inner catheter 124 in a delivery configuration in which adjustment mechanism 146 is disposed on an axis defined by sleeve 142, distally to a distal end 143 of the sleeve, and (2) transitionable into an anchoring configuration in which the adjustment mechanism is disposed laterally to the sleeve (FIG. 3A shows implant 140 in the anchoring configuration thereof). Advancement of implant 140 distally out of catheter 124 is typically controllable using a third handle 132 which is slidably coupled to handle 128, e.g., via a handle-sliding track 138.

A portion of sleeve 142 (e.g., distal end 143) is placed against annulus 10 (FIG. 3A). Typically, system 120 further comprises an implant-decoupling channel 130, disposed within sleeve 142, and slidable progressively proximally out of the sleeve, e.g., using a knob 134 coupled to a proximal end of channel 130. Typically, channel 130 is used to hold the portion of sleeve 142 (e.g., distal end 143) against annulus 10.

FIG. 3A shows anchor driver 60 (e.g., anchor-engaging head 62 thereof) being coupled to a first anchor 40 (not visible in FIG. 3A) which is disposed within a first housing 22 of device 100, as described hereinabove (e.g., with reference to FIGS. 1A-C). Subsequently, as shown in FIG. 3B, anchor driver 60 is withdrawn from the first housing 22 of device 100, while coupled to anchor 40. Anchor 40 is then advanced into system 120 by advancing driver 60 (e.g., anchor-engaging head 62 and shaft 64 thereof) through the system (FIG. 3C). Typically, and as shown in FIG. 3C, the anchor is introduced into system 120 via a port 136 at a proximal end of handle 132, and is slid through channel 130, into sleeve 142, and is screwed through sleeve 142 (e.g., distal end 143 thereof) and into annulus 10.

Driver 60 is subsequently removed from system 120, coupled to a second anchor 40 disposed in a second housing 22 of device 100, and reintroduced into the system. Channel 130 is withdrawn slightly proximally from the sleeve, and a second portion of the sleeve is held against a second site on annulus 10 before the second anchor is driven through sleeve 142, anchoring the second portion of the sleeve to the second site. This process is repeated so as to place and anchor sleeve 142 around at least a portion of annulus 10.

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Typically, and as shown in FIGS. 3A-C, device 100 is coupled to system 120 (e.g., the rest of system 120) so as to facilitate access by driver 60 to openings 28 of channels 24 of housings 22. For example, device 100 may be coupled to a stand (e.g., a base-plate) 121 of system 120, and/or may be oriented such that openings 28 face generally the same direction as port 136 of system 120, such that the operator (e.g., a physician) may easily move driver 60 between openings 28 and port 136. Typically, device 100 is positioned such that openings 28 are closer than 1 m and/or greater than 1 cm (e.g., between 1 cm and 1 m, such as between 1 cm and 70 cm, such as between 1 cm and 40 cm) away from port 136 (or another opening through which anchors 40 are introduced into system 120).

Alternatively, device 100 may comprise a standalone unit, not coupled to system 120 or any other system.

Subsequently, implant 140 may be adjusted (e.g., contracted) using an adjustment tool (not shown), advanceable over a guide member 141 to adjustment mechanism 146.

Reference is made to FIGS. 4A-F, which are schematic illustrations of an anchor-handling device 220, configured to facilitate handling of at least one tissue anchor 40, in accordance with some applications of the invention. Device 220 comprises a housing 222 that defines a channel 224 having an anchor-storage zone 226 and an opening 228 that provides access to the channel and the anchor-storage zone. Device 220 further comprises a retaining member, such as a pin 230, which is configured to retain tissue anchor 40 in zone 226, and to stop retaining the tissue anchor in response to a sufficient proximally-directed force applied to the tissue anchor. That is, when a proximally-directed force that is greater than a pre-determined threshold force is applied to tissue anchor 40, the retaining member stops retaining (e.g., releases) the tissue anchor. For some applications FIGS. 4A-F, which show steps in the use of device 220, generally correspond to FIGS. 1A-F, respectively, which show steps in the use of device 20, *mutatis mutandis*.

Typically, the retaining member (e.g., pin 230) has a retaining state in which it retains tissue anchor 40 within zone 226, and is moved out of the retaining state when the sufficient proximally-directed force is applied to the tissue anchor. FIG. 4A shows, in accordance with some applications of the invention, pin 230 in a retaining state thereof, in which at least a portion 229 (e.g., an obstructing portion) of the pin is disposed within channel 224 (e.g., proximal to anchor 40), thereby retaining the anchor in zone 226 by obstructing proximal movement of the tissue anchor. Typically, portion 229 obstructs proximal movement of anchor 40 by engaging and/or obstructing core 41 of the anchor. Another portion of pin 230 is disposed in a chamber 274, which is defined by housing 222 and is typically in fluid communication with channel 224 (e.g., in the absence of pin 230). Typically, chamber 274 has a central longitudinal axis ax1 that is parallel with a central longitudinal axis ax2 of channel 24.

Similarly to device 20, it is hypothesized that this configuring of device 220 prevents inadvertent movement and/or exit of the tissue anchor (e.g., due to general transport or handling of the device), and/or withdrawal of the anchor by driver 60 when the driver is sub-optimally coupled to the anchor.

FIG. 4A shows anchor driver 60 (described hereinabove) being advanced toward opening 228 of channel 224, and FIG. 4B shows the anchor driver having been further advanced into and through channel 224, to anchor 40, such that engaging head 62 has received anchor 40, but not yet coupled (e.g., locked) to the anchor. It is to be noted that in

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this position part of driver 60 (e.g., head 62) is disposed (e.g., sandwiched) between part of the retaining member (e.g., portion 229) and part of the anchor (e.g., coupling head 42).

FIG. 4A shows respective cross-sections of opening 228 and driver head 62, which are typically each rotationally asymmetrical. For some applications, and as shown, the cross-sections each have the shape of an ellipse (e.g., a circle) with a segment (e.g., a circular segment) removed. (For some applications opening 228 (e.g., the shape thereof) is defined partly by housing 222, and partly by pin 230.) Due to this rotational asymmetry, anchor-engaging head 62 is introducible through opening 228 in fewer than all rotational orientations of the head with respect to the opening. For example, head 62 may be introducible through opening 228 only in one or more particular rotational orientations (e.g., one particular orientation of the head) of the head with respect to the opening. This limitation of the rotational orientations in which head 62 may be introduced through opening 228 causes the head to be correctly rotationally oriented for coupling to coupling head 42 of anchor 40, the anchor being stored in zone 226 in a given rotational orientation with respect to the opening. Therefore anchor-engaging head 62 is couplable to anchor 40, without rotating the anchor-engaging head subsequently to introducing the anchor-engaging head through opening 228.

It is to be noted that this orientation-limitation may be applied to device 20, *mutatis mutandis*, and that the lateral channel opening that provides region 25 of device 20 may be applied to device 220, *mutatis mutandis*.

FIG. 4C shows wire 72 having been moved distally into engaging head 62, and detent 70 having been pushed into the closed state, thereby coupling the engaging head to anchor 40 (e.g., to coupling head 42 thereof).

FIGS. 4D-E show anchor 40 being withdrawn proximally from zone 226 by the sufficient proximally-directed force being applied to the anchor by driver 60. As described hereinabove, in response to the sufficient proximally-directed force applied to the tissue anchor (i.e., if the proximally-directed force is greater than the pre-determined threshold force), the retaining member (e.g., pin 230) stops retaining the tissue anchor in zone 226. For example, and as shown in FIGS. 4D-E, the sufficient proximally-directed force overcomes the retention provided by pin 230 such that at least a portion of pin 230 slides within chamber 274, and at least a portion 231 (e.g., a proximal portion) of pin 230 moves out of housing 222.

FIG. 4D shows portion 231 of pin 230 beginning to move out of housing 222, and FIG. 4E shows both portion 231 and anchor 40 disposed outside of the housing, such that pin 230 (e.g., portion 229 thereof) no longer obstructs anchor 40. It is to be noted that for device 220, the portion 229 of pin 230 that obstructs anchor 40 is disposed close to (e.g., within) the portion 231 of pin 230 that becomes exposed from housing 222, whereas for device 20 described hereinabove, the portion 29 of pin 30 that obstructs anchor 40 is disposed at another part (e.g., at the other end) of pin 30 from the portion 31 of pin 30 that becomes exposed from housing. Similarly, portion 229 typically becomes exposed from housing 222 upon withdrawal of anchor from device 20, whereas portion 29 typically remains within housing 22 upon withdrawal of anchor 40 from device 220.

FIG. 4F shows anchor 40 having been fully removed from the housing. For some applications, to facilitate full disengagement of anchor 40 from pin 230, the anchor is moved slightly laterally with respect to pin 30. Once anchor 40 has

been fully withdrawn, driver 60 may be used to anchor tissue anchor 40 to tissue of a subject, e.g., as described hereinabove.

Device 220 comprises an inhibitor, configured to configure the retaining member (e.g., pin 30) to (i) retain the tissue anchor in anchor-storage zone 26, and (ii) to stop retaining the tissue anchor in response to the sufficient proximally-directed force. For example, pin 230 may comprise or define a detent 282 that, while anchor 40 is disposed within anchor-storage zone 226, is held by a spring mechanism within a cavity 232 (e.g., a notch) defined in chamber 274, and thereby serves as the inhibitor. For some applications at least a portion 280 of pin 230 is resilient, and thereby provides the spring mechanism. It is to be noted, however, that the scope of the invention includes the use of other spring mechanisms. The inhibitor provides resistance that (i) inhibits sliding of pin 230 through chamber 274, e.g., prevents sliding of the pin in the absence of a sufficient proximally-directed force (e.g., as shown in FIG. 4C), and (ii) stops inhibiting the sliding in response to the sufficient proximally-directed force by detent 282 moving out of cavity 232 (e.g., as shown in FIG. 4D). For example, and as shown, resilient portion 280 deforms (e.g., bends) in response to the sufficient proximally-directed force. This is typically facilitated by detent 282 and a proximal wall of cavity 232 having respective faces that are appropriately angled with respect to each other such that the proximally-directed force is converted into lateral movement of the detent out of the cavity. For example, and as shown, detent 282 may have a beveled edge.

Typically, for such applications, once detent 282 has moved out of cavity 232, a proximally-directed force that is smaller than the threshold force is sufficient to move pin 30 further proximally. That is, once the initial resistance provided by the inhibitor is overcome, anchor 40 is further withdrawable using a smaller force than that required to overcome the initial resistance.

(It will be understood by those skilled in the art that it is possible to use other configurations to achieve a behavior similar to that described above. For example, housing 222 may define a protrusion (e.g., a detent), and pin 30 may comprise a cavity (e.g., a notch) into which the protrusion extends.)

For some applications, the inhibitor provides the resistance by applying friction against the wall of cavity 232. For example, pin 230 may have a high-friction wall-contacting surface.

Device 220 is described hereinabove with reference to only one channel 224, zone 226, and restraining member. Typically however, the device defines a plurality of channels 224, each channel having a respective proximal opening 228 and a respective anchor-storage zone 226 that is configured to store a respective tissue anchor 40 (e.g., as described with reference to device 100, mutatis mutandis). Typically, device 220 further comprises a plurality of retaining members (e.g., pins 230), each retaining member being configured to retain a respective tissue anchor in its respective anchor-storage zone 226, and to stop retaining the respective tissue anchor in response to the sufficient proximally-directed force being applied to its respective tissue anchor.

As described hereinabove, for some applications, the sufficient proximally-directed force pushes portion 231 of pin 30 out of housing 222. Therefore, as driver 60 is withdrawn proximally, movement of portion 231 out of housing 222 indicates that anchor-engaging head 62 has been successfully coupled to tissue anchor 40, and that the tissue anchor is also being withdrawn proximally. Thus,

during an initial partial withdrawal of driver 60, movement of portion 231 out of housing 222 provides an indication to the operator (e.g., physician) to continue to withdraw driver 60, whereas absence of such movement of portion 231 provides an indication to the operator to reattempt coupling of the driver to tissue anchor 40.

Following removal of anchor 40 from channel 224, portion 231 remains exposed outside of housing 222. This may be particularly useful for a physician using device 220, e.g., to prevent the physician inadvertently attempting to obtain an anchor from an empty anchor-storage zone 226. That is, portion 231 functions as an empty-housing indicator.

For some applications, a second cavity 234 is defined in chamber 274 (e.g., a second notch in a wall of the chamber), disposed more proximally with respect to housing 222 than is cavity 232. Second cavity 234 is positioned such that when (1) portion 229 is no longer obstructing anchor 40, and (2) portion 231 is exposed out of opening 278, detent 282 (i.e., the inhibitor) moves into the second cavity (e.g., as shown in FIG. 4E). In this state, the detent inhibits pin 230 from moving distally back into housing 222, thereby increasing the reliability of portion 31 functioning as an empty-housing indicator.

For some applications, and as shown in FIGS. 1A-F, second cavity 234 is dimensioned and/or shaped such that once detent 282 has moved into (e.g., engaged) second cavity 234, a distally-directed force required to return portion 231 into housing 222 (e.g., to return device 220 into its retaining state) is more than twice as great (in the opposite direction) as the threshold force that was previously required to move portion 231 out of the housing, and to remove anchor 40 from the housing. For example, and as shown, cavity 234 may be greater in one or more dimensions (e.g., wider and/or deeper) than cavity 232. While detent 282 is disposed in cavity 232, the beveled edge of the detent is partly exposed from cavity 232, whereas while the detent is disposed in cavity 234, the beveled edge is disposed entirely within the cavity. Other geometric configurations may also be used to generate this effect. For example, only a proximal face of detent 282 may be beveled.

For some applications, housing 222 is at least in part transparent, so as to enable viewing of the coupling of driver 60 to anchor 40, and/or withdrawal of the anchor from the housing.

Reference is made to FIGS. 5A-C, which are schematic illustrations of a base 300 to which device 220 (e.g., housing 222 thereof) is couplable, and which is configured to at least partly immobilize the housing, in accordance with some applications of the invention. For some applications, and as shown, more than one device 220 is couplable to base 300. Similarly to devices 20 and 100, for some applications device 220 is used with multi-component tubular system 120 (described hereinabove). For such applications, anchor driver 60 (e.g., anchor-engaging head 62 thereof) is coupled to an anchor 40, and then advanced into system 120 (e.g., via a port 136 at a proximal end of handle 132). As described hereinabove, shaft 64 is flexible, and typically has a length greater than 20 cm, and/or less than 2.5 m, such as greater than 50 cm and/or less than 1.5 m, e.g., between 0.9 m and 1.2 m. Because of this flexibility and length, it may be difficult for an operator (e.g., a physician) to wield and operate handle driver 60 (e.g., to couple head 62 to an anchor while retaining control of shaft 64 and handle 66). Base 300 facilitates such handling by defining a handle receptacle 302 for housing and at least partly immobilizing handle 66.

Typically, base 300 is configured such that when device 220 (e.g., housing 222 thereof) is coupled to the base, the

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housing is disposed less than 30 cm from receptacle 302 (e.g., less than 20 cm, e.g., less than 10 cm, such as less than 5 cm from the receptacle), and therefore less than 30 cm from handle 66 when the handle is disposed in the receptacle. Because shaft 64 is typically flexible, despite its length typically being greater than (e.g., more than twice as great, e.g., more than 5 times as great, such as 2-10 times as great as) the distance between device 220 and receptacle 302, driver head 62 is insertable into device 220 while handle 66 is disposed in receptacle 302.

For some applications, device 220 is permanently coupled to base 300 (e.g., device 220 and base 300 may be integrated). For some applications device 220 is reversibly couplable to base 300. For example, base 300 may further define at least one housing receptacle 304, configured to house device 220. For such applications, base 300 further comprises a locking element 306, movable between a locked state that locks the housing within the receptacle, and an unlocked state that facilitates release of the housing from the receptacle. For example, and as shown, locking element 306 may comprise one or more bars 308 that are advanceable through a portion of base 300 so as to protrude into (e.g., through) receptacle 304; and housing 222 is shaped to define a respective one or more recesses 310 dimensioned to mate with the bars. Advancing bars 308 into receptacle 304 while housing 222 is disposed in the receptacle thereby locks the housing within the receptacle.

Typically, receptacle 302 is dimensioned such that adjuster 68 is operable while the receptacle houses handle 66. This, along with the proximity of handle 66 to device 220, advantageously facilitates the operator (i) with a first hand, grasping a distal portion of driver 60 and introducing driver head 62 into the opening of housing 222, and (ii) with a second hand, reversibly actuating driver head 62 by operating the adjuster while grasping the distal portion of the driver with the first hand (FIG. 5C; operator's hands not shown). It is hypothesized that this advantageously improves welding and operation of driver 60 in combination with device 220.

For some applications, base 300 is coupled to system 120, e.g., as described hereinabove for device 100 with respect to FIGS. 3A-C, mutatis mutandis.

For some applications, base 300 does not define a handle receptacle, but instead serves only to hold device 220.

For some applications, device 20, device 100, and/or device 220 is used in combination with one or more techniques described in one or more of the following references, which are all incorporated herein by reference:

U.S. patent application Ser. No. 12/437,103 to Zipory et al., filed May 7, 2009, which published as US 2010/0286767. For example, (1) device 100 of the present application may be used to facilitate the techniques described with reference to FIGS. 2-3 and/or 6A-12 of US 2010/0286767 to Zipory et al., mutatis mutandis; (2) anchor driver 60 of the present application may comprise or correspond to anchor driver 68 and/or anchor deployment manipulator 24 of US 2010/0286767 to Zipory et al., mutatis mutandis; (3) tissue anchor 40 of the present application may comprise or correspond to anchor 38 of US 2010/0286767 to Zipory et al., mutatis mutandis; and/or (4) implant 140 of the present application may comprise or correspond to annuloplasty ring 22 of US 2010/0286767 to Zipory et al., mutatis mutandis.

U.S. patent application Ser. No. 12/689,635 to Zipory et al., filed Jan. 19, 2010, which published as US 2010/0280604. For example, (1) device 100 of the present

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application may be used to facilitate the techniques described with reference to FIGS. 2-3 and/or 11A-17 of US 2010/0280604 to Zipory et al., mutatis mutandis; (2) anchor driver 60 of the present application may comprise or correspond to anchor driver 68 and/or anchor deployment manipulator 24 of US 2010/0280604 to Zipory et al., mutatis mutandis; (3) tissue anchor 40 of the present application may comprise or correspond to anchor 38 of US 2010/0280604 to Zipory et al., mutatis mutandis; and/or (4) implant 140 of the present application may comprise or correspond to annuloplasty ring 22 of US 2010/0280604 to Zipory et al., mutatis mutandis.

PCT patent application IL2012/050451 to Sheps et al., filed Nov. 8, 2013, which published as WO 2013/069019. For example, (1) device 100 of the present application may be used to facilitate the techniques described with reference to FIGS. 14A-I of WO 2013/069019 to Sheps et al., mutatis mutandis; (2) system 120 of the present application may comprise or correspond to system of WO 2013/069019 to Sheps et al., mutatis mutandis; (3) anchor driver 60 of the present application may comprise or correspond to anchor deployment manipulator 61 and/or anchor driver 36 of WO 2013/069019 to Sheps et al., mutatis mutandis; and/or (4) implant 140 of the present application may comprise or correspond to annuloplasty structure 222 and/or sleeve 26 of WO 2013/069019 to Sheps et al., mutatis mutandis.

PCT patent application IL2013/050860 to Sheps et al., titled "Controlled steering functionality for implant-delivery tool", filed on Oct. 23, 2013, which published as WO 2014/064694. For example, (1) device 100 of the present application may be used to facilitate techniques described with reference to FIGS. 10A-I, 12A-14B, 18A-C, 21-28, 34, and 36 of this PCT application titled "Controlled steering functionality for implant-delivery tool", mutatis mutandis; (2) system 120 of the present application may comprise or correspond to system of this PCT application titled "Controlled steering functionality for implant-delivery tool", mutatis mutandis; anchor driver 60 of the present application may comprise or correspond to anchor deployment manipulator 61, anchor driver 36 and/or anchor driver 2338 of this PCT application titled "Controlled steering functionality for implant-delivery tool", mutatis mutandis; and/or (4) implant 140 of the present application may comprise or correspond to annuloplasty structure 222 and/or sleeve 26 of this PCT application titled "Controlled steering functionality for implant-delivery tool", mutatis mutandis.

PCT patent application IL2013/050861 to Herman et al., titled "Percutaneous tissue anchor techniques", filed on Oct. 23, 2013, which published as WO 2014/064695. For example, (1) device 100 of the present application may be used to facilitate the techniques described with reference to FIGS. 9A-C and/or 13A-D of this PCT application titled "Percutaneous tissue anchor techniques", mutatis mutandis; (2) tissue anchor 40 of the present application may comprise or correspond to tissue anchor 40 of this PCT application titled "Percutaneous tissue anchor techniques", mutatis mutandis; and/or (3) anchor driver 60 of the present application may comprise or correspond to anchor driver 500, anchor driver 236, deployment manipulator 261, or tool 80 of this PCT application titled "Percutaneous tissue anchor techniques", mutatis mutandis.

Reference is again made to FIGS. 1A-5C. Typically, the tissue anchor is dimensioned to fit snugly in the anchor-storage zone of the housing. Typically, the tissue anchor (e.g., core 41 thereof) is dimensioned to slide snugly through the channel of the housing, and for some applications this snug sliding prevents tissue-engaging member 44 of the anchor from touching the housing (e.g., the wall of the channel) when the anchor moves through the channel. Typically, at least a portion of the pin is dimensioned to slide snugly through the chamber.

As described hereinabove, for some applications the obstructing portion of the retaining member of devices 20 and 220 obstructs tissue anchor 40 by engaging core 41 of the anchor. The movement of the retaining member in response to the proximally-directed force applied to the anchor typically moves the obstructing portion such that it does not subsequently engage tissue-engaging member 44 of the anchor. For example, for device 20, portion 29 moves at least partly laterally out of the anchor-storage zone and/or channel 24 (such that member 44 can move past portion 29 without being engaged by it), and for device 220, portion 229 moves longitudinally out of channel 224 (such that member 44 can move past portion 229 without being engaged by it). It is hypothesized that for some applications this advantageously reduces a likelihood of the anchor-handling device (e.g., the obstructing portion of the retaining member) damaging tissue-engaging member 44. It is hypothesized that the use of a retaining member that has an obstructing portion that returns to its original position as soon as core 41 has moved past the obstructing portion, does not have this advantageous feature.

It is to be noted that the technique used to inhibit movement of the retaining member of device 20 may be used to inhibit movement of the retaining member of device 220, mutatis mutandis, and vice versa.

Reference is again made to FIGS. 1A-5C. It is to be noted that, for both device 20 and device 220, proximal withdrawal of anchor 40 typically results in sliding of the retaining member (e.g., the pin) in an at least partly proximal direction. This sliding is typically along an axis that is disposed at an angle of less than 30 degrees with respect to a central longitudinal axis of the channel through which the anchor is withdrawn. For example, for device 20 the angle is typically 8-30 degrees (e.g., 10-20 degrees, such as 11-14 degrees), and for device 220 the angle is typically less than 10 degrees, such as 0 degrees—i.e., parallel with the channel.

Reference is again made to FIGS. 1A-5C. As described hereinabove, the anchor-handling devices allow retrieval of the tissue anchor(s) disposed therein in response to a proximally-directed force that is greater than a threshold force. Typically the threshold force is greater than 300 grams force and/or less than 1500 grams force (e.g., 300-1500 grams force, e.g., 500-1200 grams force, e.g., 500-1000 grams force, such as 600-800 grams force). Tissue anchor 40 typically has a mass of less than 1 g and/or greater than 0.01 g (e.g., 0.01-1 g, e.g., 0.05-0.2 g, e.g., 0.07-0.12 g, such as about 0.1 g). Thus the threshold force (measured in grams force) is typically greater than 300 times (e.g., greater than 1000 times, e.g., greater than 3000 times, e.g., greater than 10,000 times, such as greater than 15,000 times) and/or less than 150,000 times the mass of the tissue anchor (measured in grams). It is to be noted that the threshold force is therefore many times greater than that which would be required simply to prevent the tissue anchor from undesirably exiting the device due to gravity and/or movement of the device (e.g., during transport).

This configuration of the anchor-handling device serves to test coupling of the anchor driver to the tissue anchor before releasing the tissue anchor. Only if the coupling is sufficient to support a proximally-directed force that is greater than the threshold force, will the device release the anchor. This is hypothesized to increase safety and reliability of the use of the anchor and driver, e.g., by reducing a likelihood that the anchor will subsequently become disengaged from the driver at an undesired time (e.g., within the body of a subject). Whereas one might consider testing the anchor-driver coupling subsequently to removal of the anchor from the anchor-handling device, such post-removal testing requires an extra procedural step, and for some applications it increases a likelihood of damaging and/or contamination of the (typically sterile) tissue anchor. Furthermore, whereas the anchor-handling devices described herein facilitate making a second attempt at coupling the driver to the same anchor, post-removal testing typically does not.

Reference is again made to FIGS. 1A-5C. Typically the anchors are provided sterile within the anchor-handling device. As described hereinabove, for some applications, the anchor-handling device is configured such that returning the exposed portion of the retaining member back into the housing requires a distally-directed force that is more than twice as great (in the opposite direction) as the threshold force that was previously required to move the portion out of the housing. For example, the moving of the portion back into the housing may be in effect prevented. As well as facilitating the exposed portion serving as an empty-housing indicator, this characteristic of the anchor-handling device discourages and/or prevents the operator from returning a previously-removed anchor into the device, e.g., thereby ensuring that only sterile anchors are disposed within the device.

For some applications, the anchor-handling devices described herein are configured to be at least in part submerged in saline prior to and/or during use, e.g., to reduce a likelihood of air (e.g., bubbles) being retained by the anchor and/or driver and subsequently introduced into the subject.

It will be appreciated by persons skilled in the art that the present invention is not limited to what has been particularly shown and described hereinabove. Rather, the scope of the present invention includes both combinations and subcombinations of the various features described hereinabove, as well as variations and modifications thereof that are not in the prior art, which would occur to persons skilled in the art upon reading the foregoing description.

The invention claimed is:

1. A system, comprising:

a tissue anchor; and
an anchor-handling device:

comprising a housing, shaped to define a channel having an anchor-storage zone and a proximal opening, the anchor stored in the anchor-storage zone, configured such that, while the tissue anchor is stored in the anchor-storage zone, the tissue anchor is movable out of the anchor-storage zone toward the proximal opening in response to a proximally-directed force being applied to the tissue anchor, and comprising an element that serves as an indicator of movement of the tissue anchor out of the anchor-storage zone toward the proximal opening by the element sliding proximally within the housing in response to the proximally-directed force:

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from an anchor-stored position that is indicative of the presence of the anchor within the anchor-storage zone,

into an anchor-absent position that is (i) different from the anchor-stored position, and (ii) indicative of an absence of the anchor from the anchor-storage zone by a part of the element protruding proximally out of the housing.

2. The system according to claim 1, wherein the system is configured to resist returning of the part of the element distally into the housing.

3. The system according to claim 1, wherein:

the tissue anchor comprises a coupling head and a tissue-engaging member,

the system further comprises an anchor driver comprising an anchor-engaging head, a handle, and a shaft therebetween, and

the anchor-engaging head is dimensioned to be advanceable into the anchor-storage zone, and configured to lock to the coupling head within the anchor-storage zone, and

the anchor driver is configured to apply the proximally-directed force to the anchor while the anchor-engaging head is locked to the coupling head within the anchor-storage zone.

4. The system according to claim 3, wherein the shaft is flexible and is configured to transluminally advance the tissue anchor into a subject while the anchor-engaging head remains locked to the tissue anchor.

5. The system according to claim 1, wherein:

the housing is shaped to define a chamber that is in fluid communication with the channel, and

the element is configured to slide within the chamber in response to the proximally-directed force applied to the tissue anchor.

6. The system according to claim 5, wherein:

the element is configured to slide proximally within the chamber in response to the proximally-directed force applied to the tissue anchor,

the anchor-handling device comprises:

a detent; and

a cavity, and

in response to movement of the element sufficiently proximally (i) to allow the tissue anchor to be removed from the anchor-handling device, and (ii) such that the element transitions into the anchor-absent position, the detent moves into the cavity, thereby inhibiting subsequent returning of the part of the element into the housing by inhibiting distal movement of the element with respect to the chamber.

7. The system according to claim 6, wherein:

the element defines the detent, and

the housing defines the cavity.

8. The system according to claim 6, wherein:

the element defines the cavity, and

the anchor-handling device further comprises an inhibitor tongue, the inhibitor tongue defining the detent.

9. The system according to claim 6, wherein:

the anchor-stored position is a retaining state in which the element is configured to retain the tissue anchor in the anchor-storage zone, and

the element is configured to allow the tissue anchor to leave the anchor-storage zone by moving in response to the proximally-directed force only if the proximally-directed force is greater than a pre-determined threshold force.

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10. The system according to claim 9, wherein:

the cavity is a first cavity of the anchor-handling device, the anchor-handling device further comprises a second cavity,

in the anchor-stored position of the element, the detent is disposed within the second cavity, the disposition of the detent within the second cavity retaining the element in the anchor-stored position by inhibiting proximal movement of the element with respect to the chamber, and

the anchor-handling device is configured such that the detent is movable proximally out of the second cavity only in response the proximally-directed force exceeding the threshold force.

11. The system according to claim 9, wherein:

the element has a distal part,

the part of the element is a proximal part of the element, and wherein:

in the anchor-stored position, the distal part engages the tissue anchor, and

in response to the proximally-directed force applied to the tissue anchor, the element slides proximally within the chamber and into the anchor-absent position, such that the proximal part of the element protrudes proximally out of the housing.

12. The system according to claim 1, wherein the tissue anchor is in contact with the element such that the proximally-directed force applied to the tissue anchor drags the element proximally, along with the tissue anchor.

13. The system according to claim 12, wherein the element defines a recess and wherein:

the tissue anchor resides within the recess, and

the system is configured such that once the tissue anchor is exposed out of the proximal opening, the tissue anchor is removable from the recess laterally with respect to the element.

14. Apparatus comprising:

a tissue anchor; and

an anchor-storage device, comprising:

a housing, shaped to define:

a cavity,

a channel having an anchor-storage zone and

a proximal opening, the anchor stored in the anchor-storage zone, and

a retaining member that:

comprises a detent,

has a retaining state in which the retaining member is configured to retain the tissue anchor in the anchor-storage zone, and

is transitionable away from the retaining state in response to a proximally-directed force applied to the tissue anchor:

to allow the tissue anchor to leave the anchor-storage zone in response to the proximally-directed force, and

such that the detent moves proximally and into the cavity to engage the cavity in a manner that inhibits subsequent movement of the retaining member distally within the housing.

15. The apparatus according to claim 14, wherein at least part of the retaining member is elastically deformable.

16. The apparatus according to claim 14 wherein:

the cavity is a first cavity,

the housing defines a second cavity, and

the retaining member is arranged such that, responsively to the proximally-directed force being applied to the tissue anchor, the retaining member deforms and moves

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proximally with the tissue anchor such that the detent moves from the first cavity into the second cavity.

17. The apparatus according to claim 16, wherein the detent is configured to, when in the second cavity, inhibit subsequent returning of the retaining member distally into the housing. 5

18. A system, comprising:
a tissue anchor; and
an anchor-handling device:
comprising a housing: 10
defining a cavity,
shaped to define a channel having an anchor-storage zone and a proximal opening, the anchor stored in the anchor-storage zone, and
comprising a retaining member that: 15
defines a detent,
engages the housing by the detent protruding into the cavity in a manner that retains the tissue anchor within the anchor-storage zone, and
responsively to a proximally-directed force being 20
applied to the tissue anchor, deforms and moves proximally with the tissue anchor to allow the anchor to move out of the anchor-storage zone and the housing.

19. The system according to claim 18, wherein at least 25 part of the retaining member is elastically deformable.

20. The system according to claim 18, wherein:
the cavity is a first cavity,
the housing defines a second cavity, and

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the retaining member is arranged such that, responsively to the proximally-directed force being applied to the tissue anchor, the retaining member deforms and moves proximally with the tissue anchor such that the detent moves from the first cavity into the second cavity.

21. The system according to claim 20, wherein the detent is configured to, when in the second cavity, inhibit subsequent returning of the retaining member distally into the housing.

22. The system according to claim 18, wherein:
the tissue anchor comprises a coupling head and a tissue-engaging member,
the system further comprises an anchor driver comprising an anchor-engaging head, a handle, and a shaft therebetween, and
the anchor-engaging head is dimensioned to be advanceable into the anchor-storage zone, and configured to lock to the coupling head within the anchor-storage zone, and
the anchor driver is configured to apply the proximally-directed force to the anchor while the anchor-engaging head is locked to the coupling head within the anchor-storage zone.

23. The system according to claim 22, wherein the shaft is flexible and is configured to transluminally advance the tissue anchor into a subject while the anchor-engaging head remains locked to the tissue anchor.

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